

NEURAL BELLMAN OPERATORS

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We develop certificate-aligned neural Bellman operators for dynamic economic decisions. A complete-cover differentiable objective guides neural proposals, while independent directed arithmetic determines acceptance. A quantitative envelope theorem links that implemented objective to policy loss, and stopping-accessibility conditions correct an inconsistent hard-settlement approximation space. On the original nonlinear economy, all ten prescribed raw certificate trajectories improve with work, although none reaches the unchanged .01 target; matched ablations and a computable witness floor identify the remaining obstruction. In a heterogeneous nonlinear inventory problem, full-state neural initialization and a provably contracting correction yield a state-uniform total-cost bound below 1.037×10^{-6} at 128 coordinates. Classical methods are compared under the same certificates and remain competitive. A separate sharp original-economy policy library admits explicit budget-financed initial-wealth compensation. The analysis distinguishes a valid certificate, improvement of a witness, actual policy accuracy, and numerical work.

KEYWORDS: Stochastic control, Neural Bellman operators, Preference adjustment, Validated numerics, Policy verification, Computational economics.

1. INTRODUCTION

Neural approximations to dynamic economic decisions require a criterion that is linked to the value of the policy actually delivered. Average equation residuals need not provide that link. A small error at frequently visited training states can coexist with a large error near a stopping face, and improving a value witness need not improve the policy with which it is paired. These distinctions become operational when an algorithm must return a complete error bound rather than a sampled diagnostic.

This paper develops certificate-aligned neural Bellman operators. The primary algorithm minimizes a differentiable envelope of two signed Bellman residuals on a complete state–time cover. An independent directed-arithmetic verifier checks the proposed policy and witness against the original economic model. Every prescribed proposal is retained; an incumbent is replaced only when its complete certificate decreases. A quantitative theorem connects the trainable envelope to the rigorous policy-loss bound, separating the smoothing allowance from a checkable numerical correction. The theorem concerns the loss actually used in the computation, rather than an unexecuted policy-iteration idealization.

Stopping geometry is part of this construction. In our consumption–investment economy with costly preference adjustment, continuation value has a strict jump exceeding 6.37 utility units at an inward upper-wealth face. A critic forced to equal immediate settlement on every face therefore cannot acquire a vanishing uniform positive residual. A signed accessibility-weighted comparison permits the critic to dominate settlement at a face that the candidate cannot reach, while retaining every originally admissible action in the optimal upper comparison. This corrects an approximation-space inconsistency without modifying preferences, controls, or the stopping contract.

The original-economy experiment compares two generations of training on that unchanged problem. Under the historical sampled objective, extending training from 800 to 2,400 updates worsens all ten certificates, even though sampled residual mean squared error falls for nine seeds.

Under the complete-cover objective, all ten accessibility-aware trajectories have decreasing raw certificates at 0, 100, and 400 updates. Their final bounds range from 7.27832 to 7.34367, compared with initial bounds from 26.3005 to 26.6668. A paired ten-seed all-face experiment, unrestricted state-space comparators with method-specific witnesses, and a fixed-network refinement study identify what causes this improvement and what it does not establish. In particular, the .01 original-economy target is unchanged and is not attained. A certified fixed-witness floor and actor–critic interchange show why neither further arithmetic precision nor a monotone incumbent alone establishes accurate policy learning.

A second, nonlinear application supplies an explicit accuracy–work result on a genuinely vector-valued state. A heterogeneous multi-product inventory planner has 12 decision dates, coupled nonquadratic costs, and up to 128 state coordinates. The neural map takes the full initial state and returns a control plan, without optimal-value or optimal-control labels. Strong convexity yields a state-uniform certificate for exact-real gradient corrections to that plan. At 128 coordinates, the total-cost bound falls from 0.124820 after 16 corrections to 0.000359609 after 24 and below 1.037×10^{-6} after 32. This is an unknown-value nonlinear planning problem, not the two-mode Riccati example retained for validation. Zero-start gradient, accelerated-gradient, and quasi-Newton alternatives are evaluated with the same independent cost-loss certificate. They remain strong competitors; the experiment establishes an accurate neural-initialized solver, not a demonstrated neural speed advantage.

Economic interpretation also requires keeping numerical objects distinct. The sharp deterministic time-control/dual library in the original economy is not a neural result. Its price-uniform regret below 0.009982 can be offset by a rigorously checked, budget-financed initial-wealth increment of .0078 at initial wealth 1.25, using an explicitly adjusted feasible consumption policy. This sufficient compensation measure respects the original consumption cap and stopping contract. The earlier externally financed top-up remains a secondary normalization. Neither measure converts the much larger original neural certificates into sharp welfare claims.

1.1. *Contribution and relation to the literature*

Stopped verification, Bellman contraction, strong-convexity error bounds, and gradient contraction are classical ingredients. Judd (1998) and Kushner and Dupuis (2001) develop numerical dynamic programming and controlled Markov-chain approximation. Barles and Souganidis (1991) establishes a fundamental monotone-approximation framework, and Brumm and Scheidegger (2017) develops adaptive sparse-grid methods for computational economics. Information and control relaxations provide dual comparisons (Brown, Smith, and Sun, 2010, Brown and Smith, 2011). Neural dynamic-control and differential-equation methods include Han and E (2016), Han, Jentzen, and E (2018); the historical external comparison uses Cai, Fang, and Zhou (2025).

Our contribution is the combination of an identified stopping-trace inconsistency, an executed complete-cover neural objective with a checkable policy-loss implication, and a nonlinear control specialization with an explicit state-uniform work–accuracy bound. The new objective theorem does not claim global convergence of Adam, and the nonlinear specialization does not claim to remove the difficulty of arbitrary stochastic high-dimensional Bellman equations. The original-economy upper comparison always retains the full control opportunity. The arithmetic implementation uses MPFR (Fousse et al., 2007); shared analytic formulas, stopping arguments, and program logic remain part of its trusted mathematical base.

The paper presents the model and verification foundations first, followed by the stopping correction, the certificate-aligned algorithm, the original-economy evidence, and the nonlinear control application. Historical coefficient transport, state–price continuation, external holdouts,

adverse training outcomes, and specialized dual calculations remain available in the main text and supplement. All new numerical statements are tied to immutable source and result objects; all 60 prescribed original-economy checkpoints are included in the new objective audit.

2. THE STOPPED PREFERENCE-ADJUSTMENT ECONOMY

Time is continuous, with horizon $T = 1$. Preferences u and wealth x evolve as

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. The state domain is

$$D = (1.2, 2.8) \times (.5, 2), \quad A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

An admissible control $a = (c, \theta, p)$ is progressively measurable, takes values in A , and is used up to the first exit from D or time one, whichever occurs first. The controls in the original optimization are not restricted to deterministic time functions or to neural networks.

For adjustment price $k \in [.5, 8]$, the running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1 - s). \quad (4)$$

Let $\tau = 1 \wedge \inf\{s \geq t : (u_s, x_s) \notin D\}$ and $\rho = .04$. The criterion is

$$J_k(t, z; a) = \mathbb{E}_{t,z} \left[\int_t^\tau e^{-\rho(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-\rho(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (5)$$

Here $V_k = \sup_{a \in \mathcal{A}} J_k(\cdot; a)$. The liquidation fee makes a stopping-trace error economically meaningful even when the training distribution assigns little mass near an exit. It is therefore included explicitly in verification rather than approximated by an average boundary loss.

For a smooth test function v , define the discounted residual

$$\begin{aligned} \mathcal{R}_a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\}v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 - .04v. \end{aligned} \quad (6)$$

All constants in the certificate denote the displayed exact decimals. Training uses floating-point approximations of them, but the certification program encloses the exact constants and treats stored binary64 network weights as exact real numbers.

The central initial condition is $(t, u, x) = (0, 2, 1.25)$. We also study the nondegenerate rectangle

$$K = [1.98, 2.02] \times [1.24, 1.26]. \quad (7)$$

The claim on K is uniform over its interior, not a statement about 72 isolated initial conditions. Its proof is given in Section 8, and its tolerance differs from the sharper central-state tolerance.

3. CONSTRUCTIVE NEURAL VERIFICATION

We first state the verification result for a class of stopped controlled diffusions. Let Z take values in a bounded domain, with drift b_a , covariance matrix Σ_a , running reward f_a , constant discount $\rho \geq 0$, and settlement g on the parabolic boundary. Write

$$\mathcal{R}_a v = v_t + b_a \cdot Dv + \frac{1}{2} \Sigma_a : D^2 v + f_a - \rho v.$$

The assumptions below concern the stochastic model and the computed witness. They do not assume regularity of the unknown optimal value.

ASSUMPTION 1: *For every admissible control, the stopped diffusion and its payoff are well defined. The controlled coefficients and payoffs are bounded on the stopped domain. The candidate feedback π is admissible. The witness v is continuously differentiable in time and twice continuously differentiable in state on a neighborhood of the closed domain, and its derivatives are bounded there. Discounted Itô's formula and optional stopping hold for these bounded stopped quantities.*

The compact economy in Section 2 with the stored smooth, bounded neural feedback satisfies these requirements. Boundedness also permits localization of the stochastic integral before taking expectations.

THEOREM 1—A complete residual and improvement certificate: *Under Assumption 1, suppose verified nonnegative functions $e(s), q(s)$ and a number $b \geq 0$ satisfy, at every state and time,*

$$|\mathcal{R}_\pi v| \leq e(s), \quad 0 \leq \sup_{a \in A} \mathcal{R}_a v - \mathcal{R}_\pi v \leq q(s), \quad |v - g| \leq b \text{ on the stopping boundary.} \quad (8)$$

Then, for every initial state z and time t ,

$$0 \leq V(t, z) - J(t, z; \pi) \leq 2b + \int_t^T e^{-\rho(s-t)} \{2e(s) + q(s)\} ds. \quad (9)$$

The same conclusion applies to any admissible implementation whose action lies in a verified output-error neighborhood of the stored feedback, provided that the residual and improvement inequalities have been enclosed over that neighborhood.

The theorem is deliberately an a posteriori statement. It is useful because every item in (8) has a finite implementation, not because its stochastic comparison identity is new. A training history can propose the witness and policy, but cannot replace any of the inequalities. The proof in Appendix A separates the policy-evaluation and upper-comparison errors; each stopping-trace discrepancy is paid once.

3.1. A finite implementation

Fix a rational rectangular cover of the entire state–time domain, including every stopping face. On each cell, interval forward differentiation gives the value, first derivatives, and relevant second derivatives of the stored network. An action oracle bounds the maximum over the continuous action set. Interval evaluation of the deployed policy gives its residual and the difference from this global maximum. Taking the largest bounds in each time slab produces e and q . Facewise

interval evaluation produces b . Finally, integration of these piecewise-constant bounds yields (9).

A cell is accepted only if all operations have finite, valid enclosures and every potential action branch is covered. No failed or inconclusive cell can be removed from the domain. Refinement replaces a cell by a complete cover of that cell, while keeping previous results in the execution record. The method either returns the certified inequality for the specified stored object or records that a requested accuracy has not been established at the available resolution. This is a stopping criterion for an accuracy claim, not a criterion based on validation loss.

In the economy, the action maximization separates into three scalar problems conditional on the interval-enclosed derivatives. Consumption is strictly concave, and its maximizer is

$$c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u}) \quad \text{when } v_x > 0, \quad c^* = .8 \quad \text{when } v_x \leq 0.$$

The preference action is $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. The portfolio objective is

$$h_1 p + \frac{1}{2} h_2 p^2, \quad h_1 = .06 x v_x - .0025 x v_{ux}, \quad h_2 = .04 x^2 v_{xx}.$$

A negative curvature gives a clipped vertex; nonnegative curvature gives an endpoint maximum. When an interval does not determine the sign, the calculation encloses the whole feasible portfolio interval. Thus uncertainty enlarges the bound instead of excluding an action. These formulas preserve the original continuous control opportunities.

3.2. Transport across model coefficients

The affine dependence on k used later is not a general argument for perturbing dynamics. For the latter we use residual transport. Consider another model with the same state domain, action set, and settlement, but possibly different coefficients and discount. Let its discount be at least $\rho \geq 0$, and let $\tilde{\mathcal{R}}_a$ denote its residual.

PROPOSITION 1—Coefficient transport: *Suppose the original checks (8) hold, and interval coefficient bounds verify*

$$\sup_{a \in A} |\tilde{\mathcal{R}}_a v - \mathcal{R}_a v| \leq d(s) \tag{10}$$

throughout the common domain. If Assumption 1 also holds for the new model, its policy regret is at most

$$2b + \int_t^T e^{-\underline{\rho}(s-t)} \{2e(s) + q(s) + 2d(s)\} ds. \tag{11}$$

If the settlement changes by at most b_{par} , replace b by $b + b_{\text{par}}$.

For example, the modulus in (10) can be evaluated directly from

$$|(\tilde{b}_a - b_a) \cdot Dv| + \frac{1}{2} |(\tilde{\Sigma}_a - \Sigma_a) : D^2 v| + |\tilde{f}_a - f_a| + |\tilde{\rho} - \rho| |v|.$$

It includes the change in the covariance matrix rather than treating volatility perturbations as drift changes. The proposition does not couple paths or equate stopping times across economies. Each model is verified against its own stopped dynamics on the common domain. The executed coefficient box is reported in Section 14.

4. STOPPING ACCESSIBILITY AND THE NEURAL APPROXIMATION SPACE

A stopping contract need not be the limit of continuation value at every geometrical boundary point. This distinction matters for a degenerate controlled diffusion: an investor can eliminate portfolio risk and move away from a wealth boundary. It therefore cannot be decided by assigning a larger penalty to every boundary observation in a neural loss.

PROPOSITION 2—Trace consistency in the unchanged economy: *For the economy in Section 2, at $k = 2$,*

$$\liminf_{x \uparrow 2} V_2(0, 2, x) - g(0, 2, 2) \geq 6.45 - .1 \log 2 - 25.1e^{-8} > 6.37. \quad (12)$$

Consequently, a sequence of witnesses continuous on the closed cylinder, each equal to g on its entire stopping boundary, cannot have its uniform positive Bellman residual converge to zero.

The proposition concerns the consistency of an approximation space, not the existence of a solution to the economic decision problem. The original utility, adapted controls, horizon, and liquidation fee are unchanged. Its implication is constructive: enforce the settlement where the proposed policy can stop, and use the correct one-sided comparison where it cannot. In particular, imposing the upper-wealth settlement as a hard equality would exclude the relevant continuation limit. The proof uses the elementary feasible policy $(c, \theta, p) = (.8, 0, 0)$ and does not use an estimated optimal value.

4.1. A signed, facewise comparison

Partition the stopping boundary into measurable faces F_i , assigning intersections to one face. Let

$$\omega_i^*(t, z) \geq \sup_a \mathbb{E}_{t,z} [e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)], \quad \omega_i^\pi(t, z) \geq \mathbb{E}_{t,z}^\pi [e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)].$$

These are upper bounds on discounted hitting probabilities, not probabilities estimated by simulation. A nonnegative smooth function whose discounted generator is nonpositive and whose stopping trace dominates the face indicator supplies such a bound by optional stopping. A separate invariance argument can give the exact value zero for an inaccessible face.

THEOREM 2—Accessibility-weighted neural verification: *Under Assumption 1, suppose complete enclosures establish*

$$\sup_a \mathcal{R}_a v \leq e_+(s), \quad -\mathcal{R}_\pi v \leq e_-(s), \quad e_+, e_- \geq 0,$$

and $(g - v)_+ \leq \beta_i^+$, $(v - g)_+ \leq \beta_i^-$ on F_i . For an interior initial state,

$$\begin{aligned} 0 \leq V(t, z) - J(t, z; \pi) &\leq \int_t^T e^{-\rho(s-t)} \{e_+(s) + e_-(s)\} ds \\ &\quad + \sum_i \beta_i^+ \omega_i^*(t, z) + \sum_i \beta_i^- \omega_i^\pi(t, z). \end{aligned} \quad (13)$$

The upper comparison retains the entire original action set. Inaccessibility for the candidate policy does not remove an action from that comparison.

This theorem retains the stopped verification identity underlying Theorem 1. The new distinction is which sign of a trace discrepancy must be paid, and under whose stopping distribution it is paid. Replacing both signs by a common supremum discards precisely the boundary information identified by Proposition 2. The inequality is still an a posteriori certificate; convergence of a candidate-generation procedure is a separate statement.

4.2. An executable accessibility-preserving architecture

Let N_c, N_θ, N_p, N_v be smooth neural outputs. We use

$$\begin{aligned} c &= .425 + .375 \tanh N_c, & \theta &= .2 \tanh N_\theta, \\ p &= \frac{2-x}{1.5} \{ .15 + .65 \tanh N_p \}, \end{aligned} \tag{14}$$

$$v = G + (1-t) \{ -8 + B(u, x) \text{softplus}(N_v) \}, \tag{15}$$

$$B(u, x) = (1 - e^{-12(u-1.2)})(1 - e^{-12(2.8-u)})(1 - e^{-4(x-.5)}).$$

Here $\text{softplus}(y) = \log(1 + e^y)$. Consumption and preference controls obey the original bounds. The portfolio head also obeys the original bounds throughout the state domain. Its boundary factor is part of the policy, not a replacement of the economy's portfolio opportunity.

PROPOSITION 3—Exact elimination of the trace budget: *For an interior initial state, the policy in (14) cannot hit $x = 2$ before another stopping event. The witness in (15) equals g on the two preference faces, the lower wealth face, and the terminal face, and is at least g on the upper wealth face. Hence every trace contribution in (13) is zero.*

The statement remains true for bounded admissible perturbations of the action heads, clipped to their stated intervals *before* multiplication of the portfolio head by $(2-x)/1.5$. An unrestricted additive error in the final portfolio output is different: it can destroy inaccessibility and is not covered by this claim. The replication records this implementation contract explicitly. The certificate describes a real mathematical feedback with stored dyadic weights; it is not a claim about an undocumented floating-point activation routine or an SDE simulator.

The architecture does not by itself make the residual small. Full-domain derivative enclosures must still validate both terms in (13). Reporting a zero trace budget without the remaining residual budget would therefore be incomplete. Nor do we force the numerical continuation witness to equal the known immediate-settlement value at an initially stopped upper-boundary state: at that state every policy has value g and regret zero directly.

5. AN AUXILIARY FINITE-BELLMAN POLICY-ITERATION BOUND

The following result is an auxiliary statement about a stationary finite Bellman problem, not the algorithm executed in the nonlinear experiments. The executed certificate-aligned algorithm is defined in Section 6. It is useful to distinguish three sources of approximation: policy evaluation, policy improvement, and approximation of a continuous-time transition problem. We state a finite Bellman result that makes the first two requirements quantitative. It does not assume that a particular stochastic-gradient optimizer attains them.

Consider a finite discounted Markov problem with rewards r_a , stochastic or substochastic transition matrices P_a , and contraction modulus $q \in (0, 1)$. Absorbing settlement is incorporated

in the reward. Write $T_a v = r_a + qP_a v$ and $Tv = \max_a T_a v$. A neural evaluation v_n of policy π_n and an improved policy π_{n+1} are accepted with recorded defects

$$\|v_n - T_{\pi_n} v_n\|_\infty \leq \epsilon_n, \quad 0 \leq Tv_n - T_{\pi_{n+1}} v_n \leq \eta_n \mathbf{1}. \quad (16)$$

On a continuous state space the same inequalities require a complete enclosure, not a finite collection of collocation residuals. On a finite state space they can be checked by enumeration. They give a stopping and refinement criterion for a learned representation without identifying its training loss with its Bellman defect.

THEOREM 3—A quantitative certified policy-iteration recurrence: *Let $E_n = \|V^* - V^{\pi_n}\|_\infty$ for the finite Bellman problem. If (16) holds, then*

$$E_{n+1} \leq qE_n + \frac{2q\epsilon_n}{(1-q)^2} + \frac{\eta_n}{1-q}. \quad (17)$$

Thus geometrically decreasing certified defects imply convergence of policy values, with the explicit convolution bound obtained by iterating (17). If both defects are bounded by Cr^n , $r \in (0, 1)$, the bound decreases at rate $O(\max\{q, r\}^n)$ when $q \neq r$, and $O(nq^n)$ when $q = r$.

The contraction and policy-improvement ingredients are classical. The operational content is the use of independently checked evaluation and continuous-action defects as acceptance criteria. A neural optimizer that does not achieve those defects has not met the hypothesis. In particular, the full-horizon experiments below are not assigned a convergence guarantee by virtue of being called policy iteration.

A time or state discretization requires an additional comparison. If a numerical optimal value differs from continuous-time value by at most ζ_h , and the numerical value of the transferred policy differs from its continuous-time payoff by at most ξ_h , its continuous-time loss is at most $E_n + \zeta_h + \xi_h$. Neither term is assumed zero. An alternative is to bypass both transfers and certify the actual continuous-time feedback with Theorem 2; that is the procedure used for the classical candidates below. At an inward stopping face, imposing an inconsistent Dirichlet approximation does not supply the required ζ_h . Proposition 2 explains why this qualification is material rather than technical.

For a fixed positive discount rate, $q = e^{-\rho h}$ approaches one as h decreases. The factors in (17) must therefore be included in any accuracy allocation. The theorem is not a dimension-free complexity result. The original nonlinear economy still uses a full rectangular certificate; the structured extension in Section 11 avoids a state tensor cover for a different, explicitly proved reason.

6. A CERTIFICATE-ALIGNED NEURAL BELLMAN ALGORITHM

The primary algorithm has three explicit objects: a policy π_θ , a smooth witness v_ϕ , and a complete-cover certificate. Its target is the original continuous-time policy loss, not the optimal value of a discretized training problem. The architecture in Section 4 supplies admissibility and a zero signed trace budget. The following construction addresses the remaining residuals.

Partition time into $0 = t_0 < \dots < t_N = T$ and cover the spatial domain on slab j by M_j closed boxes. Write

$$m_j = \int_{t_j}^{t_{j+1}} e^{-\rho s} ds, \quad b_{ji}^+ \geq \sup_{(s,z) \in Q_{ji,a}} \mathcal{R}_a v_\phi(s, z), \quad b_{ji}^- \geq \sup_{(s,z) \in Q_{ji}} (-\mathcal{R}_{\pi_\theta} v_\phi)(s, z).$$

Each $b_{ji}^\pm$ is nonnegative; the optimal comparison ranges over the original continuous action set. Define

$$B(b) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \max\{0, \max_i b_{ji}^\sigma\}, \quad (18)$$

$$S_\tau(a) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \tau \log \left(1 + \sum_{i=1}^{M_j} \exp(a_{ji}^\sigma / \tau) \right), \quad \tau > 0. \quad (19)$$

The extra unit in each logarithm is the zero logit. It is necessary because only the positive part of each signed residual is charged. All bounds in this section are for time zero; replacing m_j by the appropriate discounted remaining-slab masses gives the corresponding bound at another initial time.

THEOREM 4—Trainable envelope and certified policy loss: *Suppose the hypotheses of Theorem 2 hold, the signed trace budget is zero, and the boxes cover the entire state–time domain. For any finite proposal logits a_{ji}^σ , let*

$$d_j^\sigma \geq \max_i (b_{ji}^\sigma - \max\{a_{ji}^\sigma, 0\})_+, \quad D = \sum_j m_j (d_j^+ + d_j^-).$$

Then every interior initial state satisfies

$$0 \leq V(0, z) - J(0, z; \pi_\theta) \leq B(b) \leq S_\tau(a) + D. \quad (20)$$

If the certified endpoints themselves are used as logits, then

$$B(b) \leq S_\tau(b) \leq B(b) + 2\tau \sum_j m_j \log(M_j + 1). \quad (21)$$

A directed upper enclosure of $S_\tau(a) + D$, computed from the exact stored proposal logits and the certified endpoints, is therefore a sufficient, checkable stopping criterion for the policy-loss target.

The theorem applies to the implemented differentiable interval-envelope loss even when its ordinary floating-point arithmetic has not itself been proved enclosing. The correction D covers any endpoint discrepancy between that proposal graph and the independently evaluated certificate; it is not assumed to be a rounding error of a particular library routine. In the new audit, the logits are recomputed from the stored neural weights and then interpreted as exact dyadic numbers for the directed soft-maximum calculation. Neither differentiation through an ordinary floating-point interval graph nor a small sampled loss is called a certificate.

The smoothing allowance is explicit. With four time slabs, 256 spatial boxes per slab, $\rho = .04$, $T = 1$, and $\tau = .15$, its upper bound is 1.631868. This is a worst-case allowance, not an observed policy loss. Refining the cover while keeping τ fixed need not remove this allowance. A sufficient joint refinement rule is $\tau \log(1 + \max_j M_j) \rightarrow 0$, together with consistent residual enclosures. The theorem does not assert that a nonconvex optimizer reaches a requested envelope level.

6.1. The executed algorithm

The complete-cover neural algorithm uses the following rule. Initialize the actor and critic, form the fixed cover, and minimize $S_\tau(a(\theta, \phi))$ by joint Adam updates of their parameters. At each predeclared checkpoint, freeze and serialize both networks. Independently recompute every signed residual box with directed arithmetic and the original action maximization. Reject an incomplete, nonfinite, or failed enclosure. For a complete object, compare its bound B with the incumbent bound, retain the proposal regardless of the comparison, and replace the incumbent only under a strict decrease. Stop at the declared work budget or when the complete bound meets the stated target. The implementation is `guided_neural.py`; the separate checker is `accessibility_certificate.py`.

The reported original-economy run uses checkpoints 0, 100, 400, temperature .15, learning rate .003, gradient clipping at 100, and two hidden tanh layers of width 16 or 32. There is no point sampling in its envelope objective: all 1,024 boxes enter every proposal update. The ten seeds and paired all-face ablations use matched initialization logic, optimization budget, cover, and arithmetic. The acceptance decision uses the complete MPFR bound, not the smooth proposal loss. The new R18 audit applies Theorem 4 to these already executed, predeclared R17 trajectories; it is not represented as a newly randomized training study.

PROPOSITION 4—What certificate acceptance guarantees: *The incumbent bound of the preceding algorithm is nonincreasing and remains a valid upper bound for its associated incumbent policy. Neither this property nor a decrease in raw proposal bounds implies that the true values of successive policies increase.*

The distinction is substantive. A better witness can certify the same policy more sharply. Conversely, a worse policy can have a smaller bound if it is paired with a much better witness. Accordingly, the numerical record reports raw proposals, incumbent decisions, and actor–critic interchanges separately. A claim of true policy improvement would require an additional policy-value comparison, not just the incumbent rule.

6.2. A finite-horizon error allocation without a stationary resolvent

The auxiliary stationary recurrence in Section 5 has a different role. Its amplification factors become large when $q = e^{-\rho h} \uparrow 1$. For a finite horizon, a backward error allocation is more appropriate when a time-discrete implementation is analyzed directly. Let T_n and $T_n^{\pi_n}$ be monotone Bellman operators with Lipschitz modulus q , and let v_N approximate the terminal reward g .

PROPOSITION 5—Finite-horizon evaluation and improvement defects: *If*

$$\|v_N - g\|_\infty \leq b, \quad \|v_n - T_n^{\pi_n} v_{n+1}\|_\infty \leq \epsilon_n, \quad 0 \leq T_n v_{n+1} - T_n^{\pi_n} v_{n+1} \leq \eta_n \mathbf{1},$$

then the nonstationary policy $\pi = (\pi_0, \dots, \pi_{N-1})$ satisfies

$$\|V_0^* - V_0^\pi\|_\infty \leq 2q^N b + \sum_{n=0}^{N-1} q^n (2\epsilon_n + \eta_n). \quad (22)$$

For $q = e^{-\rho h}$, $Nh = T$, and $\epsilon_n \leq \bar{\epsilon} C_\rho(h)$, $\eta_n \leq \bar{\eta} C_\rho(h)$, the accumulated defect term is at most $(2\bar{\epsilon} + \bar{\eta}) C_\rho(T)$.

The finite-horizon statement is representation independent. Its 80 exact-rational finite-MDP regression cases do not constitute execution of a continuous-time neural policy-iteration scheme. The original-economy algorithm instead applies Theorem 4 directly to the continuous-time policy, avoiding an unproved discretization-transfer bound. Both finite-Bellman results are retained, with their different hypotheses and numerical roles made explicit.

7. A SHARP ECONOMIC COMPARISON CONSTRUCTION

A neural certificate should be compared with independently identified bounds for the same economy. We therefore retain the affine-preference market-deflator construction and reimplement its numerical evaluation. The lower policy and the upper comparison object are identified separately from the neural feedback throughout.

7.1. Feasible policies and the original stopping contract

A library policy consists of 16 exact stored consumption constants $c_j \in [.7, .8]$, preference actions $\theta_j \in [0, .2]$, and $p_j = 0$. These controls are held on their time intervals until the original first exit. Wealth is deterministic and satisfies

$$x_1 = e^{.02} \left(x_0 - \int_0^1 e^{-.02s} c_s ds \right).$$

The program proves $x_1 > .5$ and the complete action bounds before evaluating a payoff. It does not round a budget residual to zero or assume that a nearly feasible policy is feasible.

The previous implementation evaluated a polynomial representation with Gaussian moments. The new evaluator instead applies an interval-certified Gauss rule directly to the original CRRA expression on a covered Gaussian domain. After $t = v^2$, it integrates over v and a standard-normal coordinate z . Rationally isolated Gauss nodes, interval weights, and analytic complex-neighborhood remainder bounds enclose the integrals. The unbounded normal tail and the discrepancy from stopping are accounted for analytically. The use of an auxiliary bounded full-horizon extension in this proof does not change the objective (5): it is connected to that objective by an explicit stopping correction, less than 3.87×10^{-18} on K for these policies.

At $k = 4.25$, the second implementation encloses the policy payoff in

$$[-1.315469674845588, -1.315469674834266], \quad (23)$$

with the stored unrounded interval width approximately 1.13×10^{-11} . This is a policy-payoff width, not an optimality gap. The latter still requires an upper bound on V_k .

7.2. The affine-preference dual

Let $dY_s = .02Y_s ds - .3Y_s dZ_s^x$. Its product with wealth satisfies

$$d(Y_s x_s) = (.04Y_s x_s - Y_s c_s) ds + \text{a local martingale.}$$

The cancellation holds for every original adapted portfolio control. Define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad \beta = \partial_u \Psi.$$

For a deterministic reference $\bar{\theta} \in [0, .2]$, set $m(t) = 2 + \int_0^t \bar{\theta}_s ds$ and let Q be the affine tangent to $\phi(u) = -.02(u - 2)^2$ at $m(1)$. On $u \in [1.5, 2.5]$, $y \in [.2, 12]$, the supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (24)$$

It is justified by analytic monotonicity and complete interval covers, not by sampled tangency or presumed global concavity. Its new primitive checks have strictly positive margins; the supplement gives the proof and the recorded numerical margins.

Put $\hat{y} = \text{proj}_{[.2, 12]} y$ and replace the source by the right side of (24) at $\hat{y}$. The resulting full-line auxiliary problem $A_k(t, u, y)$ retains adapted preference actions, quadratic adjustment cost, and terminal payoff Q . Its source and terminal value are affine in the initial preference. The following positive localization potentials reconnect it to the original first-exit problem:

$$Z_u(h, u) = 8h\{e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h}\}, \quad (25)$$

$$Z_y(h, y) = 8h\{e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h}\}. \quad (26)$$

For $y_0 \in [1.2, 2.4]$ the resulting upper bound is

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (27)$$

The auxiliary value has a computable upper bound consisting of one-Gaussian source expectations, a terminal tangent, a correlated-shock correction, a reference-action gap, and a conditional-variance allowance. The latter is necessary because the adapted marginal value of adjustment is random; replacing it by its unconditional mean without a correction would not be valid. The supplement states and proves the complete formula, including the factors .00375 and .09/(24k) in the covariance and variance terms.

In the second implementation, the source uses separate analytic formulas for its capped and interior consumption branches. Interval automatic differentiation supplies Hessian enclosures, and a rational-root Gauss construction encloses the weighted moments. This differs from the inherited manually differentiated source and polynomial moment integrator. The analytic dual theorem is shared; its mathematical validity is not established merely by agreement between programs. Its finite supporting-plane checks and all decisive numerical quantities are recomputed by the new implementation.

8. JOINT INITIAL-STATE AND PRICE CERTIFICATION

The state-domain extension is obtained without assuming a Lipschitz modulus for the unknown optimal value. Two structural properties suffice: a convex upper comparison function and a concave feasible-policy lower function.

8.1. Extending the dual in the initial state

For the fixed dual witness, define

$$b_0 = \int_0^1 e^{-\rho s} \mathbb{E} \beta(m(s), \hat{Y}_s) ds + e^{-\rho} Q'. \quad (28)$$

An interval for b_0 is obtained in the same independent execution as the dual. Since the coefficient of the initial u in the auxiliary criterion is independent of the chosen adapted preference action,

$$A_k(0, u, y_0) = A_k(0, 2, y_0) + (u - 2)b_0.$$

Let U_0 be a certified upper endpoint in (27). It follows that

$$\begin{aligned} V_k(0, u, x) &\leq U_0 + y_0(x - 1.25) + (u - 2)b_0 \\ &\quad + 8e^{-18}\{e^{120(u-2)} + e^{-120(u-2)} - 2\}, \end{aligned} \quad (29)$$

for the verified initial rectangle. Interval multiplication is sign-aware when $u - 2 < 0$. With an interval for b_0 , its upper support function is convex in u . Thus the computable right side of (29) is convex in (u, x) . This is a property of the upper comparison object, not an assertion that V_k itself is convex in wealth.

8.2. A budget-preserving policy map

Write $Q_r = \int_0^1 e^{-.02s} ds$. For each stored policy, define

$$c_j(x) = c_j + \frac{x - 1.25}{Q_r}, \quad \theta_j(x) = \theta_j, \quad p_j(x) = 0. \quad (30)$$

Its terminal wealth is identical to that of the central-state policy. All shifted consumption values remain in $[\cdot, \cdot]$ on K ; this is checked for every library member. The policy depends on the initial state and is thereafter a time control until exit. It is not described as an approximate neural feedback surface.

The utility $f(c, u) = c^{1-u}/(1-u)$ is jointly concave over the region needed by the Gaussian cover. With $r = u - 1$ its Hessian satisfies

$$\begin{aligned} f_{cc} &= -uc^{-u-1}, \\ f_{cu} &= -(\log c)c^{-u}, \\ f_{uu} &= -\frac{c^{-r}}{r^3}\{(r \log c + 1)^2 + 1\}, \\ \det D^2 f &= \frac{c^{-2u}}{r^3}\{r^2(\log c)^2 + 2r(r+1)\log c + 2(r+1)\}. \end{aligned}$$

On $u \in [1.2, 2.8]$, $c \in [\cdot, \cdot]$, one has $r(-\log c) < 1$, so the determinant is positive. The covered expected payoff, with the exact budget shift, is therefore concave in the initial state. A single uniform tail and stopping correction turns this covered functional into a lower bound for the original stopped policy throughout K .

PROPOSITION 6—Four vertices and a continuum of prices: *Suppose, at each price node k_i , a convex upper function $U_i(z)$ bounds $V_{k_i}(0, z)$ on a rectangle K . For each policy j , suppose a concave lower function $F_j(z; k)$ bounds its original payoff on K , is affine in k , and has a verified expenditure slope interval. Then corner values alone produce a uniform regret certificate on $K \times [k_1, k_N]$ by the following finite construction. Between adjacent price nodes use the chord of U_i and U_{i+1} . For each policy transfer its four lower corner values affinely in price, using the appropriate expenditure endpoint. At each price select the policy minimizing the largest of its four upper-minus-lower corner gaps. The supremum of the resulting bound is attained at an endpoint or an intersection of the finitely many affine corner-gap functions.*

The optimal value is convex in k because the policy set, dynamics, and stopping contract are common and fixed-policy payoffs have the form $A(\pi) - kB(\pi)$. This justifies the upper

price chord, not interpolation of V as though it were affine. The expenditure used for the concave covered lower functional is its full-horizon deterministic expenditure, with the stopping discrepancy already paid uniformly at the largest price. For the separate central-state exact-policy envelope, the program instead bounds the stopped expenditure. Confusing these two slopes would invalidate the transfer; they are separately recomputed and labeled in the evidence.

PROPOSITION 7—Executed state–price certificate: *For the unchanged economy (5), the 18 stored policies with the state map (30) and the price selector in Proposition 6 satisfy*

$$\sup_{(u,x) \in K} \sup_{k \in [5,8]} \{V_k(0, u, x) - J_k(0, u, x; \pi_{k,x})\} \leq 0.012095431554. \quad (31)$$

The central-state selector based on the independently evaluated stopped policies satisfies

$$\sup_{k \in [5,8]} \{V_k(0, 2, 1.25) - J_k(0, 2, 1.25; \pi_k)\} \leq 0.009969007711. \quad (32)$$

The execution evaluates all 72 policy–state corners, uses all 18 independently verified upper witnesses, and checks 211 exact candidate breakpoints for the joint state–price construction. No unsampled state requires an empirical smoothness extrapolation. The exact rational worst-case joint bound is

$$\frac{2187960974729572475442514300375}{180891518006739738096917284913152}.$$

The displayed decimal in (31) is rounded upward. The computational record contains the complete selected-policy regions and every corner interval. Appendix B proves the interpolation and finite-breakpoint steps.

9. CERTIFICATE-ALIGNED COMPUTATION ON THE ORIGINAL ECONOMY

All experiments in this section use the original economy, its full continuous action comparison, $k = 2$, and the complete state–time domain. They are not price-library or inventory results. The immutable R17 scientific object contains the prescribed trajectories and all raw certificates. The new R18 calculations audit their objective implications, reproduce two full-domain certificates, and reconstruct every displayed aggregate from the retained records.

9.1. Raw trajectories, acceptance, and the training objective

Table I reports every seed. All ten accessibility-aware raw sequences decrease strictly at both transitions; all paired all-face sequences also decrease. Thus the improvement is not created by selecting the incumbent or hiding a rejected proposal. All 40 post-initialization proposals are accepted by the strict rule. The historical sampled-training experiment is different: all ten of its 800-to-2,400-update certificates deteriorate, although sampled residual mean squared error falls in nine of ten cases. The new training loss reverses that particular anti-alignment in the observed experiment; it is not a proof that every longer Adam run must improve the certificate.

The complete-cover objective audit covers 60 neural objects and 61,440 cells. It verifies all 589 files in the immutable scientific manifest, checks network/certificate identities and complete cell coverage, and re-evaluates the proposal logits. Directed evaluation of the soft maximum and its endpoint correction verifies Theorem 4 for every object. The largest integrated correction D is below 1.20×10^{-13} ; the remaining differences between a smooth objective and a box

TABLE I
ALL PRESCRIBED ORIGINAL-ECONOMY NEURAL TRAJECTORIES

Seed	Width	B_0	B_{100}	B_{400}	All-face B_{400}	Seconds
17100	16	26.300514	10.104267	7.339624	7.345561	104.1
17101	32	26.499830	7.547734	7.281385	7.263538	193.4
17102	16	26.326374	10.237435	7.343312	7.351242	96.1
17103	32	26.511217	7.535185	7.280303	7.263378	143.4
17104	16	26.367488	10.591643	7.336433	7.345015	71.3
17105	32	26.468295	7.555938	7.280018	7.261518	114.4
17106	16	26.352908	9.701133	7.343665	7.349999	51.2
17107	32	26.666728	7.604406	7.278319	7.262408	97.2
17108	16	26.339971	9.745279	7.338341	7.345879	48.6
17109	32	26.527472	7.553444	7.279140	7.261511	79.0

All entries use the full original state–time cylinder and continuous action upper comparison. B_j denotes a raw proposal certificate, not an incumbent replacement. Seconds include all three certificate checks and the complete accessible training trajectory. Both architectures use identical declared budgets. Printed values are rounded; exact directed bounds, signed decompositions, and all 60 objects are retained.

maximum are not an arithmetic precision effect. The audit is deliberately separate from the training experiment and retains the exact recomputed logits.

The final accessibility-aware bounds are between 7.2783190 and 7.3436651, and none reaches .01. The original target is not relaxed. The result establishes a reproducible improvement in the certified objective, not sharp accuracy of the original nonlinear neural policy. Training, verification, and checkpoint selection costs are included separately in the retained resource records. At the final checkpoint, the observed mean total trajectory cost is 99.87 seconds; timings describe the recorded hardware and concurrent workload, not a matched-accuracy speed ranking.

9.2. A matched boundary ablation and a fixed-witness floor

At width 16, the accessibility-aware final bound is smaller than its paired all-face bound in all five pairs. At width 32, the ranking reverses in all five pairs. The paired mean difference (accessibility-aware minus all-face) is approximately .005049. This is not evidence of a uniform finite-budget advantage for the corrected trace class. The structural result in Proposition 2 remains an asymptotic consistency statement, while the actual finite-budget comparison depends on the learned witness and optimization trajectory.

For seed 17100, refining a fixed final accessibility-aware object from 1,024 to 8,192 and 65,536 cells changes its bound from 7.339625 to 7.280222 and 7.251560. The corresponding all-face bounds are 7.345561, 7.267777, and 7.236151. These are verifier-refinement curves around fixed objects, not new policy training. They quantify enclosure dependence without suggesting that a sufficiently fine cover would reach .01.

PROPOSITION 8—A computable fixed-witness obstruction: *Let v satisfy the upper-comparison trace conditions and let $h_v = v(0, 2, 2) - g(0, 2, 2)$ be its upper-wealth trace excess. Write $\Delta = 6.45 - .1 \log 2 - 25.1e^{-8}$. Any nonnegative, time-dependent bound on its positive optimal residual must satisfy*

$$\int_0^1 e^{-.04s} e_+(s) ds \geq (\Delta - h_v)_+. \quad (33)$$

TABLE II
ORIGINAL-ACTION CLASSICAL CANDIDATES WITH METHOD-SPECIFIC WITNESSES

Method	Nodes/axis	Time steps	Bound	Generate	Witness	Verify
MC	9	64	7.339815	0.34	62.14	9.41
MC	17	128	7.314510	7.60	62.41	9.71
MC	25	256	7.763463	124.01	39.84	6.71
SL	9	64	7.342250	0.35	67.46	8.60
SL	17	128	7.314559	8.52	45.80	7.58
SL	25	256	7.774123	74.71	39.22	5.34

Times are seconds. MC: controlled Markov chain; SL: semi-Lagrangian. Interior actions use no neural distance multiplier. Every transferred policy has a separate witness and a complete continuous-time check; all six fail the unchanged .01 target. These bounds do not rank true policy values. Discrete central values remain diagnostics.

This is a lower bound on the certificate component of this fixed witness, not a lower bound on the regret of its policy.

The final accessible critics have verified trace excesses only about .052–.097. Their lower floors from (33) range from 6.274882 to 6.320490. Consequently, refining their covers or increasing arithmetic precision cannot close the original accuracy gap. A materially different continuation witness is required. The all-face class has the stronger structural obstruction $h_v = 0$.

The actor–critic interchange sharpens this diagnosis. Pairing the initial actor with the final critic for seed 17100 gives exactly the same retained bound, 7.339624178645005, as the final actor/final critic pair. Pairing the final actor with the initial critic gives 23.718685694842687. Thus the certificate descent cannot be attributed to a demonstrated increase in the actor’s true payoff. Both facts are retained rather than reinterpreted as policy improvement.

9.3. Unrestricted candidate generation and independent witnesses

The new controlled Markov-chain and semi-Lagrangian candidates do not multiply their interior portfolio choices by the neural wealth-distance factor. Each solves over the original bounded action grid, including zero portfolio. The upper-wealth grid column represents the interior continuation trace: zero-portfolio continuation is computed and compared with immediate settlement for nonzero portfolio at that trace. The retained policies select zero portfolio on this column. The certified controls are the original-action-clipped, exact-real bilinear state interpolants, constant within each time interval. Their interpolation consequently has bounded $p/(2 - x)$ and cannot hit the upper face before another stopping event; no interior distance multiplier is imposed.

Each candidate receives a separately trained witness and a full continuous-time residual check against the original continuous action set. Hence neither the discrete central value nor a common inaccurate neural witness is used as its accuracy certificate. Table II gives all six resulting bounds and separates candidate generation, witness construction, and verification. The method-specific witnesses remove the earlier common-witness confound but do not produce a sharp matched-accuracy frontier. In particular, the finest-grid bounds increase because their policy residual contribution is no longer zero. The numerical values of the discrete schemes remain diagnostics, not certified continuous-time values.

9.4. Control changes and numerical scope

The supplement also reports whole-domain ranges for all six classical policies and whole-price/time ranges for all twenty fresh-library policies. For the latter, consumption lies in

[.7618225, .7721747], preference adjustment in [.0031022, .2], and portfolio is exactly zero. These library ranges hold for every policy selected over $k \in [.5, 8]$, not only the displayed price nodes.

Complete-box bounds on the original neural controls are reported in the supplement for all ten final actors. For seed 17100, the final ranges are approximately [.530998, .749371] for consumption, [.093966, .113275] for preference adjustment, and [0, .087417] for portfolio. The corresponding uniform bounds on the change from initialization are .218738, .019170, and .100661. These are conservative ranges of mathematical control functions over the full state–time domain, not simulated confidence intervals or policy-optimality claims.

The certificate interprets stored weights as exact dyadic numbers and activations as mathematical tanh/softplus. The allowed head perturbation is 2^{-24} and precedes the wealth-distance factor. Neither the proposal graph’s ordinary floating-point activations, a deployed inference engine, nor an SDE simulator is certified by this contract. Independent arithmetic reduces rounding risk but does not independently establish every shared analytical premise.

10. NONLINEAR MULTI-PRODUCT CONTROL WITH A STATE-UNIFORM CERTIFICATE

A high-dimensional test should require the policy to depend on the high-dimensional state. We therefore distinguish the following nonlinear planning problem from the two-mode quadratic inventory model retained in Section 11. Here there is no Riccati reduction or supplied optimal solution.

There are d heterogeneous products and $H = 12$ dates. Initial inventory is $x \in [-1, 1]^d$. The deterministic dynamics and total cost are

$$z_0 = x, \quad z_{h+1} = \frac{3}{5}z_h + u_h - b_h, \quad b_{hi} = \frac{((h+i) \bmod 7) - 3}{100}, \quad (34)$$

$$F_x(U) = \sum_{h=0}^{H-1} \{2\|u_h\|_2^2 + \Phi(z_{h+1})\}, \quad U = (u_0, \dots, u_{H-1}) \in \mathbb{R}^{Hd}, \quad (35)$$

$$\Phi(z) = \sum_{i=0}^{d-1} \left\{ \frac{q_i z_i^2}{2} + \frac{1}{2} \log \cosh z_i \right\} + \frac{3}{10} \sum_{i=0}^{d-1} \log \cosh(z_i - z_{i+1}), \quad q_i = 1 + \frac{i}{5(d-1)}, \quad (36)$$

where indices in the coupling term are cyclic. Production/adjustment costs are quadratic, while individual and cross-product inventory penalties are nonquadratic. Every coordinate enters the nonlinear coupling; heterogeneity prevents the two-invariant-mode decomposition of the quadratic comparison model. Its Bellman equation is

$$W_h(z) = \inf_{u \in \mathbb{R}^d} \{2\|u\|_2^2 + \Phi(\frac{3}{5}z + u - b_h) + W_{h+1}(\frac{3}{5}z + u - b_h)\}, \quad W_H = 0.$$

The computation instead uses the equivalent deterministic control-plan formulation, without treating either formulation’s unknown value as training data.

A two-hidden-layer neural map takes all d coordinates of x and produces all Hd controls:

$$U_0(x) = \frac{1}{4} \tanh N_\theta(x).$$

It is trained on the original cost, not optimal labels, using 400 Adam updates, batches of 32 uniformly drawn initial states, learning rate .003, width $\min\{64, 2d\}$, and three predeclared

seeds. A correction stage applies

$$U_{k+1}(x) = U_k(x) - \frac{16}{209} \nabla_U F_x(U_k(x)). \quad (37)$$

The complete solver is the neural initialization together with the declared number of corrections. The correction, not a change of model or an optimal-value label, supplies its quantitative convergence mechanism.

THEOREM 5—Uniform nonlinear control accuracy and work: *For every $d \geq 2$ and every x , the cost in (35) satisfies*

$$4I \preceq \nabla_U^2 F_x(U) \preceq \frac{177}{8} I, \quad U \in \mathbb{R}^{Hd}. \quad (38)$$

Let $r = 145/209$. For every $x \in [-1, 1]^d$ and every bounded neural initialization above, the exact-real correction obeys

$$0 \leq F_x(U_k(x)) - \inf_U F_x(U) \leq \frac{\|\nabla F_x(U_k(x))\|_2^2}{8} \leq \frac{Hd(8.85)^2}{8} r^{2k}. \quad (39)$$

A gradient evaluation requires $O(Hd)$ arithmetic operations using forward state propagation and backward adjoints. Thus, after initialization, a sufficient number of correction steps for total loss at most ε is

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(Hd(8.85)^2/(8\varepsilon))}{2 \log(209/145)} \right\rceil \right\}.$$

No tensor cover of the initial-state cube is required.

The work statement counts exact-real arithmetic operations, not bit complexity or the cost of floating-point certification. The result uses strong convexity and the sparse coupled dynamics of this application. It is not a theorem for a general stochastic high-dimensional HJB equation. The factor d in total loss is visible, and no per-coordinate normalization is used to hide it.

10.1. Uniform and pointwise numerical objects

Table III reports the directed evaluation of the state-uniform bound for every specified dimension and correction budget. The $d = 128$ total bound is below .000360 after 24 corrections and below 1.037×10^{-6} after 32. These inequalities hold for the exact-real neural map and correction at every initial state in the cube, regardless of which of the three stored neural seeds is used.

Separately, the actual stored floating-point plans are checked by MPFR at two fixed initial vectors: an alternating corner and a predeclared pseudorandom vector. The checker evaluates the original cost and KKT gradient using exact rational model constants, and applies the first inequality in (39). It does not assume that the floating-point optimizer has converged. All 90 neural query/budget objects and 66 classical query/budget objects are retained across dimensions. At $d = 128$ and 32 corrections, the six retained neural query-plan bounds range from 8.28771×10^{-15} to 1.62604×10^{-14} . Those sharper values concern those six exact stored plans, not every state in the cube. The new R18 execution rechecks all 18 final neural query plans across the three dimensions.

TABLE III
STATE-UNIFORM TOTAL-COST LOSS FOR NONLINEAR CONTROL

Corrections	$d = 8$	$d = 32$	$d = 128$
0	939.87	3759.48	15037.9
8	2.70779	10.8312	43.3247
16	0.00780122	0.0312049	0.12482
24	2.24755×10^{-5}	8.99022×10^{-5}	0.000359609
32	6.47527×10^{-8}	2.59011×10^{-7}	1.03604×10^{-6}

The domain is every initial $x \in [-1, 1]^d$. Bounds concern the exact-real neural initialization and rational-step gradient correction, not an uncertified floating-point inference graph. Entries are total cost losses, not per-coordinate losses. The theorem holds for every bounded initialization, including an untrained one.

TABLE IV
MATCHED CERTIFICATE COMPARISONS AT 128 COORDINATES

Method	Largest query bound	Train	Online	Verify
Neural + gradient	1.63×10^{-14}	12.709	0.078	0.678
Zero + gradient	2.06×10^{-14}	0.000	0.078	0.555
Accelerated gradient	2.6×10^{-16}	0.000	0.071	0.648
L-BFGS	3.13×10^{-14}	0.000	0.096	0.699

All query plans use the same MPFR KKT-gradient certificate. Gradient methods use 32 corrections; L-BFGS uses its retained stopping result. Neural entries summarize six query/seed plans; classical entries summarize two queries. Times are descriptive means in seconds; neural training is a one-time per-network cost, not a per-query charge. Query bounds do not replace the state-uniform bound in Table III.

10.2. Matched certificate comparisons

The comparison methods are zero-start gradient descent with the same step, strongly-convex accelerated gradient, and L-BFGS. All use the same original model and the same independent gradient-based cost-loss inequality. Table IV reports the final $d = 128$ query results and separates neural training, online solution, and verification time. The supplement includes the corresponding tables at $d = 8$ and 32 and every retained correction budget.

The data do not establish a neural speed advantage: strong classical methods attain comparable or smaller certified errors without the training cost. The state-uniform theorem likewise holds for any initialization in the specified bounded class, including an untrained one. Neural initialization and its amortized cost are therefore reported as components of the solver, not credited with the contraction supplied by the correction. This application establishes an accurately certified, full-state-dependent nonlinear control-plan solver; it does not replace the original stochastic economy or close that economy's remaining neural accuracy gap.

11. RETAINED QUADRATIC INVENTORY VALIDATION

For historical validation of the dynamic certificate, retain the quadratic inventory-adjustment economy below. Its numerical gain problem reduces to two scalar Riccati equations; the genuinely vector-state nonlinear experiment is Section 10. With $M = \mathbf{1}\mathbf{1}'/d$,

$$dX_s = (AX_s + a_s)ds + .1dW_s, \quad A = -.2I + .3M, \quad (40)$$

and minimize

$$\mathcal{J}(t, x; a) = \mathbb{E} \left[\int_t^1 \{X'_s(I + .5M)X_s + |a_s|^2\} ds + .5|X_1|^2 \right]. \quad (41)$$

Controls are square-integrable adapted adjustments. Mean inventory affects drift and aggregate holding cost; idiosyncratic inventory also has its own adjustment dynamics and shocks. This is a coupled d -state dynamic problem, not an action maximization at frozen derivatives. It is a structured, uncalibrated application, and the structure is available to classical solvers as well.

The orthogonal and mean modes have $(a_\perp, q_\perp) = (-.2, 1)$ and $(a_m, q_m) = (.1, 1.5)$. Their Riccati gains satisfy

$$(p_i^*)' + 2a_i p_i^* - (p_i^*)^2 + q_i = 0, \quad p_i^*(1) = .5.$$

We train two neural gain outputs $\hat{p}_i(t) = .5 + (1 - t) \text{softplus}(N_i(t))$ against these *differential residuals*, not against exact Riccati solutions. The feedback is

$$\hat{a}(t, x) = -\hat{P}(t)x, \quad \hat{P} = \hat{p}_\perp(I - M) + \hat{p}_m M.$$

The network changes its gain with time and controls all d state coordinates. One learned pair is reused across the six dimensions; this reuse is not counted as six independent training runs.

THEOREM 6—State-global quadratic neural policy certificate: *Suppose interval differentiation on a complete time cover bounds the two neural Riccati residuals by ε_i . Put $\kappa_\perp = 1.4$, $\kappa_m = .8$, and*

$$D = \max_i \varepsilon_i \frac{1 - e^{-\kappa_i}}{\kappa_i}, \quad C = \frac{1 - e^{-.8}}{.8}.$$

For every $d \geq 1$ and every $|x|^2/d \leq 1$,

$$0 \leq \frac{\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x)}{d} \leq D^2 \left\{ C + \frac{.01}{.8} (1 - C) \right\}. \quad (42)$$

The bound concerns the continuous-time controlled diffusion on all of $\mathbb{R}^d$, not a finite collection of initial states.

The proof uses a quadratic performance-difference identity and a closed-loop Lyapunov moment bound. It replaces state covering by algebra valid on the entire state space. The certificate’s time-enclosure work depends on network size and the number of time cells, rather than a tensor grid in d . Applying the feedback requires a mean and coordinatewise operations, hence $O(d)$ work. This advantage comes from the model’s two invariant modes; it is not a generic escape from dimensional dependence for nonlinear Bellman equations.

Ten fixed seeds, two widths, and both predeclared checkpoints are retained. Every final checkpoint has per-coordinate certified loss at most 0.000095606; the complete range and costs appear in Table V. The certificate scales to total loss by multiplication by d . The supplement also reports the earlier checkpoint and fixed-network time-cover refinements separately. A structure-aware classical calculation solves only two scalar Riccati equations. Its low cost is reported rather than hidden: this application establishes accurate, reproducibly generated neural feedback with full dynamic certification, not superiority over the classical solution of this specially structured problem.

TABLE V
CERTIFIED NEURAL POLICY LOSS PER INVENTORY COORDINATE

Seed	Width	250 updates	1,000 updates	Train (s)	Verify (s)
16200	8	0.00117290	0.00009561	5.28	1.18
16201	16	0.00045470	0.00007661	7.05	2.85
16202	8	0.00051798	0.00006084	5.40	1.27
16203	16	0.00011929	0.00003574	7.44	3.26
16204	8	0.00179439	0.00008973	5.09	1.09
16205	16	0.00054105	0.00002530	6.28	3.05
16206	8	0.00036825	0.00004136	5.59	1.36
16207	16	0.00023854	0.00009053	6.34	3.34
16208	8	0.00099196	0.00005105	5.88	1.60
16209	16	0.00036174	0.00007427	5.08	2.29

Each checkpoint uses a complete 512-cell time cover and a state-global quadratic proof. Bounds hold for every $|x_0|^2/d \leq 1$. Multiply the reported bound by d for total loss at $d = 4, 8, 16, 32, 64, 128$. One gain pair is reused across dimensions. Timings cover one training trajectory and its two checks, not six duplicated runs.

11.1. A rigorously validated classical comparison for the quadratic model

The preceding quadratic model has only two independent gain equations. We retain it as a transparent validation case, not as the nonlinear high-dimensional benchmark. Its classical comparison is now certified rather than merely an RK45 diagnostic.

For each scalar mode, with coefficients $(a, q) = (-.2, 1)$ or $(.1, 1.5)$ and terminal gain $p(0) = .5$ in remaining time h , let $\kappa = \sqrt{a^2 + q}$ and $y = (.5 - a)/\kappa$. The exact gain is

$$p^*(h) = a + \kappa \frac{y + \tanh(\kappa h)}{1 + y \tanh(\kappa h)}.$$

It is increasing on $[0, 1]$. Directed endpoint values therefore enclose its full range on each time cell. The implemented policy stores a dyadic midpoint gain on each cell; MPFR bounds its uniform gain error δ . All stored gains are strictly above $.5$, so the moment estimate used in the quadratic performance-difference identity gives per-coordinate loss at most

$$\delta^2 \left\{ C_{.8}(1) + \frac{.01}{.8}(1 - C_{.8}(1)) \right\}.$$

Total loss is this quantity multiplied by d . Table VI reports both units and includes the full generation-and-verification cost. In particular, the 128-cell classical policy has a $d = 128$ total bound below $.002457$, and the 2,048-cell policy has a total bound below 9.624×10^{-6} . This is the appropriate rigorous comparison for a problem whose exact structure is so favorable to classical methods.

12. HISTORICAL SAMPLED-TRAINING AND RESTRICTED-BASELINE EXPERIMENTS

This section retains the R16 experiment without substituting later results for its observations. The original nonlinear economy is unchanged in all experiments in this section. The architecture is (14)–(15). Seeds 16100–16109 use two hidden tanh layers, with width 16 for even seeds and width 32 for odd seeds. Every trajectory retains the 800- and 2,400-update checkpoints. Each actor update follows three critic updates; Adam learning rates are $.0002$ and $.0005$, respectively, with batch size 128, double precision, and one Torch thread. A quarter of each sampling mixture

TABLE VI
VALIDATED CLASSICAL POLICIES FOR THE TWO-MODE QUADRATIC MODEL

Time cells	Per-coordinate bound	Total bound, $d = 128$	Seconds
32	0.000304119	0.0389272	0.064
128	1.91901×10^{-5}	0.00245634	0.127
512	1.20221×10^{-6}	0.000153883	0.380
2048	7.51824×10^{-8}	9.62334×10^{-6}	0.929

Complete time-cell range enclosures use the closed-form scalar Riccati solutions and MPFR-directed arithmetic. The policies are dyadic piecewise-constant gains. Both generation and validation are included in seconds. This is the structured quadratic comparison, not the nonlinear application.

targets preference and lower-wealth boundary layers. No optimal-value labels are used. These settings were fixed before the canonical remote execution; no best-seed substitution is made.

For every checkpoint, MPFR interval differentiation encloses the complete $[0, 1] \times [1.2, 2.8] \times [.5, 2]$ cylinder on $4 \times 16 \times 16$ cells. The continuous consumption, adjustment, and portfolio maxima use the original action set. The certificate allows a componentwise head perturbation of 2^{-24} , clipped to the head bounds before the portfolio-distance multiplication. This location of the perturbation matters: an unrestricted additive error in the final portfolio could invalidate the boundary-inaccessibility argument. The proof concerns mathematical real-valued networks with stored dyadic weights under this explicit output contract. It does not certify a particular machine-library tanh implementation or an SDE simulation stack.

TABLE VII
FULL-HORIZON NEURAL CHECKPOINTS IN THE ORIGINAL ECONOMY

Seed	Width	800 updates	2,400 updates	Train (s)	Verify (s)
16100	16	30.19468	54.00405	165.70	28.36
16101	32	41.45535	96.87386	180.11	79.46
16102	16	28.84397	53.60464	156.26	29.22
16103	32	42.60784	93.28879	162.35	59.47
16104	16	29.69505	68.44466	138.88	23.06
16105	32	48.37134	100.68512	139.71	65.74
16106	16	30.03684	59.19474	129.16	27.60
16107	32	45.64637	102.35557	143.36	62.72
16108	16	28.54732	49.67500	132.95	25.88
16109	32	44.11682	95.77234	74.49	23.67

Each bound is uniform over the original interior state rectangle at time zero, $k = 2$, and uses 1,024 cells with the original continuous action maximum. The trace contribution is exactly zero. Training time covers both checkpoints in one trajectory; verification time sums their two audits. No checkpoint meets 10^{-2} . Resource measurements are descriptive, not matched-accuracy efficiency comparisons.

Table VII retains every prescribed checkpoint. The exact trace budget is zero, but the positive optimal residual and negative policy residual still produce large bounds. An increase in optimization budget uniformly loosens these certificates. Since the bounds also include interval dependency, this observation does not by itself prove that the realized policy payoff deteriorates. The refinement experiment holds the first seed’s final network fixed and doubles the cover in each coordinate; the complete cell records separately retain the signed budget and the more conservative unsigned residual/action budget. The hard-all-face ablation is retained as well.

Its finite-resolution certificate is not evidence that the excluded approximation space becomes consistent as residual error tends to zero.

12.1. State-space candidate generators

We implement a controlled Markov-chain approximation with nonnegative transition weights and a semi-Lagrangian weak Euler approximation with correlated shocks. Each uses the same utility, discount, action bounds, state domain, and first-exit settlement. Refinement uses 9, 17, and 25 nodes per state coordinate, 64, 128, and 256 time steps, and progressively richer bounded control-head grids. The Markov-chain implementation uses wide diffusion jumps and a checked nonnegative stay probability; the semi-Lagrangian method uses four symmetric shock directions. Approximate trajectories apply the liquidation contract at their first straight-segment exit. This discretization is a candidate-generation device, not an asserted exact solution of the stopped diffusion.

For verification, each numerical policy is extended by bilinear interpolation in state and piecewise-constant interpolation in time. Its portfolio head is multiplied by $(2 - x)/1.5$, exactly as in the neural candidate class. These are feasible policies in the unchanged economy, although the candidate class is narrower than the original collection of all admissible controls. The upper comparison still uses every original control. We enclose the interpolated policies with the same complete verifier and the fixed seed 16100 final critic. Thus their discrete value outputs are never used as continuous-time upper or lower bounds.

TABLE VIII
HISTORICAL RESTRICTED STATE-SPACE CANDIDATES WITH A COMMON WITNESS

Method	Nodes/axis	Time steps	Regret bound	Generate (s)	Verify (s)
Markov chain	9	64	48.98350	0.12	3.84
Markov chain	17	128	48.95819	3.41	3.73
Markov chain	25	256	49.45469	42.90	3.74
Semi-Lagrangian	9	64	48.93352	0.08	3.67
Semi-Lagrangian	17	128	49.47658	1.06	3.69
Semi-Lagrangian	25	256	49.69852	25.90	3.74

All six policies are continuous bilinear state interpolants of bounded heads, piecewise constant in time, with the portfolio-distance factor. The common witness is the predeclared seed 16100 final critic, not a fitted value from either baseline. Generation discretizes the model; the separate certificate evaluates the continuous-time policy and unchanged stopping contract. None meets 10^{-2} , so this is not a matched-accuracy ranking.

The common witness standardizes the verification contract but is inaccurate; these loose bounds are not a comparison of solver accuracy. None of these six candidates attains the 10^{-2} target under that certificate. The experiment supplies two distinct state-space constructions, three refinements, and actual continuous-time policy verification; it does not establish a matched- 10^{-2} efficiency frontier or an optimal implementation of either classical method. In particular, a loose common upper witness cannot rank realized policy quality merely by the size of a certified upper bound. Section 9 reports the later method-specific witnesses. Their remaining error is not an omitted term that can be declared negligible.

13. ARITHMETIC INDEPENDENCE AND FRESH PRICE-CONTINUUM CONSTRUCTION

We distinguish two complete calculations. The first re-evaluates all eighteen historical policy/dual nodes with a new arithmetic implementation. Every basic operation and every expo-

nential, logarithm, square root, and tanh endpoint is evaluated using 128-bit MPFR directed rounding, followed by directed conversion to binary64 endpoints. Stored network weights and policy constants are interpreted as exact dyadic inputs, while model decimals are enclosed as exact rationals. Invalid domains and division through zero fail closed. The kernel passes 450 exact-rational regression checks, but these checks are diagnostics rather than a formal proof of the implementation.

The full-node MPFR calculation recomputes stopped policy payoffs, dual source integrals, expenditure intervals, and the final exact rational envelope. It does not read historical certified endpoints as inputs. It does share the R14 analytic decomposition, rational Gauss-node isolation, derivative remainder lemmas, and stopping inequalities. Thus it is an independent established-arithmetic cross-check of the decisive calculation, not an independent discovery of every mathematical premise. Its complete price-uniform bound is 0.009969007708; full lower and upper intervals, not three pointwise comparisons, are retained at every node.

The second calculation starts from the economic model alone. It initializes the price set at $1/2$ and 8, generates a 16-slab feasible time-control policy at each node by deterministic SLSQP initialization, and fits the corresponding dual from deterministic starts. It then selects the worst exact current envelope breakpoint, rounded to a spacing of $1/1024$; a repeated node triggers bisection of the widest price interval. The declared stopping rule is a proved uniform gap below .01 or thirty freshly generated nodes. No inherited policy, dual pilot, or historically selected intermediate node is read. Optimizer attempts, unsuccessful intermediate envelopes, and resource use are retained.

TABLE IX
COMPLETE PRICE-CONTINUUM CALCULATIONS ON THE ORIGINAL ECONOMY

Calculation	Nodes	Uniform regret	Wall time (s)	Peak RSS (MiB)
Historical policies; MPFR	18	0.009969007708	1065.89	276.3
Fresh model-to-library	20	0.009981645276	415.79	497.5

Both cover $(t, u, x) = (0, 2, 1.25)$ and every $k \in [.5, 8]$. Historical inputs are frozen policies and dual pilots, not inherited endpoints; MPFR recomputes every node. The fresh calculation starts from model constants and deterministic initializations, with no historical policy or pilot. Its arithmetic is the retained rational–Taylor implementation; the MPFR label does not apply to the fresh library. Peak MPFR RSS is the largest worker, not summed concurrent memory.

The fresh pipeline reaches 0.009981645276 with 20 generated nodes. This result belongs to the time-control/dual method, not to the neural feedback. The fresh calculation uses the retained rational–Taylor interval kernel; the independent MPFR label belongs only to the separately re-evaluated eighteen-node object. Fresh generation closes a reproducibility distinction without removing the dual relaxation’s accuracy floor. The original 16/32/64-slab experiment and its failure to attain .005 are retained below. Neither extra price nodes nor quadrature refinement alone supplies a convergence law to arbitrarily small error.

The historical R16 replication command executes that revision’s neural trajectories and checks, both state-space solvers, the fresh price library, the historical-node MPFR audit, the structured inventory experiment, and the normalization calculations. Source identities, input hashes, complete cell records, failures, and hardware/software metadata accompany the results. Timings in different concurrent experiment processes are descriptive; they are not used to infer solver superiority from unequal accuracy or unequal processor contention.

TABLE X
WHOLE-DOMAIN NEURAL CERTIFICATES ON THE UNCHANGED ECONOMY, $k = 2$

Updates	Cells	$2b$	$\int e^{-\rho s}(2e + q)ds$	Regret upper	Seconds
100	2048	14.9778	8.8713	23.8491	4.40
300	2048	14.2970	9.0431	23.3400	4.75
800	256	14.2484	25.9010	40.1494	0.42
800	2048	14.1073	16.1408	30.2480	4.41
800	16384	14.0463	9.4470	23.4932	25.09

Upper bounds are rounded upward. Time is the measured verification stage, excluding training. Every row covers the complete domain with zero skipped and zero failed cells. The last three rows refine the cover for the same 800-update snapshot. None attains the .01 target.

13.0.0.1. *Historical evidence retained.* The following original-economy experiments, bounds, comparisons, and welfare-resolution calculations are the R14 research record. Their sources and results are unchanged. They are not additional R16 training runs or endpoints of the fresh R16 library.

14. RETAINED R14: EXECUTED NEURAL VERIFICATION ON THE ORIGINAL ECONOMY

The retained R14 neural experiment uses a state-feedback actor and critic with two hidden layers of 48 tanh units. Inputs are $(2t - 1, (u - 2)/.8, (x - 1.25)/.75)$. The actor's output transformation is

$$c = .425 + .375 \tanh a_1, \quad \theta = .2 \tanh a_2, \quad p = .15 + .65 \tanh a_3.$$

The critic is $v(t, u, x) = G(u, x) + (1 - t)N(t, u, x)$, so its terminal trace is exact. The first-exit trace is not imposed by this architecture and must be checked.

All weights are freshly generated from seed 1401. Each actor-improvement step is preceded by three critic residual updates, with batch size 128. Adam learning rates are .0005 for the critic and .0002 for the actor, and the boundary-loss coefficient is .025. Snapshots at 100, 300, and 800 updates are retained. The experiment uses the 48-by-48 tanh architecture of the retained comparisons, but its continuous economic residual and boundary objective are a new adapter; it is not the same finite-step training objective as the external benchmark. The optimizer and sampled objective are proposal mechanisms only.

The certification program uses an independent interval implementation for network evaluation and differentiation. It checks all cells of $[0, 1] \times [1.2, 2.8] \times [.5, 2]$, all four first-exit faces, and the terminal face. A componentwise output neighborhood of radius 2^{-24} is included in the action ranges. This proves robustness for any admissible action within that neighborhood of the mathematical real-tanh network. It does not assert an undocumented error bound for a machine-library tanh routine or for a discretized diffusion simulator.

Table X separates the stopping trace from the integrated policy residual and action gap. The finest cover has 16,384 interior cells and 512 cells on each stopping face. It gives the time-zero bound 23.493172463 and an all-initial-times bound below 23.656877772. The terminal trace is zero. The 800-update sampled residual mean square is approximately .0079446, but this does not control either the worst-case residual or the first-exit trace. Moreover, the 800-update snapshot is not uniformly better than the 300-update snapshot at the same coarse cover. Both outcomes remain in the record.

The coefficient-transport calculation uses the same stored snapshot and cover, with

$$r \in [.019, .021], \quad \mu \in [.058, .062], \quad \sigma_u \in [.049, .051], \quad \sigma_x \in [.195, .205],$$

TABLE XI
STOPPING-TRACE ERRORS FOR THE FINEST NEURAL COVER

Stopping face	Cells	Trace error upper
$u = 1.2$	512	4.63971373
$u = 2.8$	512	7.02049755
$x = .5$	512	5.17780198
$x = 2$	512	7.02310795

Each entry is a uniform bound over the complete face and time interval. The comparison theorem pays the largest face bound twice. The error is an evaluation of the stored critic against $G - 8(1 - t)$, not a change in that settlement.

$$\rho \in [.039, .041], \quad k \in [1.9, 2.1],$$

and correlation fixed at $-.25$. Here μ denotes the risky-wealth drift coefficient multiplying p . The action bounds, state domain, and settlement are unchanged. Proposition 1 gives a time-zero regret bound below 23.530407814 uniformly over this model box. This demonstrates an executed transport mechanism beyond affine objective continuation. It does not make the base neural certificate accurate: the same boundary and approximation errors remain visible.

The finest computation records separate cellwise evaluation, action, and transport quantities. A diagnostic comparison of the interval neural derivatives with automatic differentiation checks 140 derivative components at 20 states and 60 actor components, with no discrepancy outside the numerical tolerance. This diagnostic detects implementation disagreement; it is not substituted for the full-cell interval proof.

15. RETAINED R14: INDEPENDENT NUMERICAL EVIDENCE AND ROBUSTNESS

The sharp economic comparison uses a second implementation that imports none of the inherited primal or dual certifiers. Its basic arithmetic encloses exact rational constants and exact stored binary64 inputs. Addition, multiplication, division, and reductions are rounded outward. Exponentials and logarithms are bounded by rational series with analytic remainders; $\tanh$ is obtained from the enclosed exponential. The Gauss nodes are isolated by exact polynomial signs and the weights are enclosed by their defining formula. The supplement states the floating-point model and the analytic remainder bounds, so these assertions are not based solely on rerunning a production function.

The independent primal evaluator uses 262,144 integrand evaluations per 16-interval policy. The fine dual uses 262,144 source boxes per price. Complete supporting-plane covers certify a negative second-derivative upper bound below $-.06717$ on the first preference region and a strict derivative separation on the remaining region. They also certify a source floor above -6.98956 , a source ceiling below zero, and an absolute preference coefficient below $.97119$. These independently computed margins imply the analytic bounds $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ used for stopping continuation. The complete old policy and dual inputs are retained and identified by their content hashes.

Exact price postprocessing is independently implemented by enumerating affine intersections with rational arithmetic, rather than importing the original hull routine. Its input expenditures are newly computed from the stored preference actions with an explicit stopping correction. Thus the upstream policy values, upper values, expenditure intervals, and downstream envelope all have a new execution path. Reproducibility of this path remains distinct from independence of its analytic premises.

TABLE XII
INDEPENDENT CENTRAL-STATE CERTIFICATES FOR ALL RETAINED PRICES

k	Policy lower	Optimum upper	Regret upper	Policy width
0.5	-1.27266301	-1.26281817	0.009844825	1.13e-11
0.875	-1.27897331	-1.26925551	0.009717794	1.13e-11
1.25	-1.28479324	-1.27510918	0.009684054	1.13e-11
1.625	-1.29013398	-1.28045672	0.009677256	1.13e-11
2	-1.29501788	-1.28533017	0.009687701	1.13e-11
2.5	-1.30083961	-1.29112854	0.009711056	1.13e-11
3	-1.30590538	-1.29616492	0.009740453	1.13e-11
3.5	-1.31024866	-1.30047467	0.009773977	1.13e-11
4	-1.31389776	-1.30408950	0.009808244	1.13e-11
4.25	-1.31546968	-1.30564683	0.009822839	1.13e-11
4.5	-1.31688907	-1.30705332	0.009835742	1.13e-11
5	-1.31935183	-1.30949330	0.009858515	1.13e-11
5.5	-1.32141556	-1.31153758	0.009877966	1.13e-11
6	-1.32317048	-1.31327568	0.009894785	1.13e-11
6.5	-1.32468139	-1.31477191	0.009909467	1.14e-11
7	-1.32599609	-1.31607368	0.009922399	1.14e-11
7.5	-1.32715060	-1.31721673	0.009933858	1.14e-11
8	-1.32817262	-1.31822851	0.009944109	1.14e-11

Every upper endpoint covers all original adapted controls; every lower endpoint belongs to the identified feasible policy. The last column is the width of that policy's payoff enclosure, not its regret. Regret uses unrounded endpoints. All 18 cases are included, and no inherited generation time is counted as zero.

TABLE XIII
SEPARATE QUANTITIES IN THE INDEPENDENT $k = 4.25$ CALCULATION

Quantity	Lower	Upper
Source integral	-2.38605836997	-2.38604817428
Covariance	-0.00017879144	-0.00016944433
Reference-control gap	0	0.00003256016
Conditional variance	0.00003697298	0.00003697299
Localization	2.4594E-7	2.4595E-7
Primal payoff	-1.315469674846	-1.315469674834

Intervals are displayed outward. The source-remainder enclosure, Gaussian tail, terminal tangent, reference cost, and arithmetic allowances are present in the machine-readable record. The conditional-variance allowance is positive and is not removed when quadrature is refined.

15.1. Stress tests for the entire library

The robustness experiment changes every dual pilot: y_0 is increased by 10^{-6} , and its reference preference drifts receive alternating perturbations of $\pm 10^{-6}$ with inward projection into the admissible reference interval. Simultaneously, the proof uses the weaker source floor -7.11 , the larger coefficient bound 1.11 , twice the Gaussian-tail allowance, and 1.1 times the conditional-variance allowance. The resulting continuation floor remains strictly above -8 . All 18 upper certificates are recomputed, while the original economy and feasible lower policies remain unchanged.

The unperturbed independent bound leaves approximately 3.10×10^{-5} below .01, compared with approximately 5.98×10^{-6} for the retained R12 bound. The perturbed calculation remains

TABLE XIV
CONTINUUM CERTIFICATES UNDER INDEPENDENT VERIFICATION AND PERTURBATION

Certificate	Uniform regret upper
R12 retained implementation, original inputs	0.009994024847
R14 independent implementation, finer cover	0.009969007711
R14 perturbed witnesses and enlarged allowances	0.009970977450
R14 joint state–cost rectangle	0.012095431554

The first three rows refer to the central initial state. The fourth covers the additional initial-state rectangle K and is not compared with the central-state .01 tolerance as though the domains were identical. The perturbation experiment changes proof witnesses and allowances, not the economic primitives.

below .01. Neither statement claims a bound below .009. An additional exact sensitivity calculation shows how a common enlargement of every upper and lower endpoint changes the final envelope: enlarging each side by δ increases the guaranteed gap by at most 2δ . It is a sensitivity result conditional on valid node intervals, not a replacement for the independent primitive audit.

A third, materially different decomposition integrates the capped and interior conjugate using closed truncated-lognormal moments. At prices .5, 4.25, and 8, a separate program computes the source integral, initial affine coefficient, and covariance term at 40 and 70 decimal digits. Every computed quantity lies inside the independently certified interval. The agreement provides a useful cross-check of algebra and decomposition; these point-valued high-precision quadratures are not themselves rigorous enclosures, and their convergence is not used as a proof of the interval certificate.

16. RETAINED R14: ACCURACY, COMPUTATIONAL WORK, AND COMPARATORS

A certification time measured from frozen proof objects is not the cost of generating those objects. The 18-node independent audit takes approximately 103.08 seconds of measured wall time in the retained run, but starts from existing policies and dual pilots. The 72-corner state audit takes approximately 29.71 seconds. These are verification costs. The generation costs of inherited and interrupted historical attempts are not reconstructed from their absence and are not assigned zero.

The fresh neural run records all 800 actor updates and 2,400 critic updates, the fixed seed, every retained checkpoint, and resource usage. Its recorded execution after program entry is approximately 13.08 seconds, with interpreter and import time separately recorded. It uses one Torch CPU thread and no GPU. The execution environment reports an AMD EPYC 9V74 processor and five exposed logical CPUs; this identifies the environment, not exclusive ownership of a physical machine. Stage times recorded during overlapping local experiments are descriptive clocks, not controlled hardware-efficiency comparisons.

16.1. *A newly generated policy-richness frontier*

The same original economy at $k = 2$ is also solved within deterministic time-control classes of 16, 32, and 64 intervals. Each problem starts from a fixed feasible constant-consumption construction and preference action .05. Constrained sequential quadratic optimization uses analytic gradients to propose a policy. A separately initialized scalar-deflator and reference-drift fit proposes its upper witness. The fresh policy is then checked by the independent stopped evaluator and the fresh upper witness by the independent dual. The fitting objective is not counted as a certificate.

TABLE XV
FRESH GENERATION AND INDEPENDENT VERIFICATION AS POLICY RICHNESS INCREASES

Slabs	Generation	Dual fit	Primal audit	Dual audit	Regret upper
16	0.235	0.284	0.737	7.408	0.009688058
32	0.364	0.434	0.926	6.815	0.009669548
64	0.847	0.379	1.791	5.748	0.009665074

All stage times are seconds. Each case starts without inherited policy or dual inputs, and all three optimizations report success. End-to-end measured case times are 8.67, 8.54, and 8.77 seconds. The experiment changes policy richness, unlike a price-node refinement experiment. All cases pass .01; none passes .005, .0025, or .001.

The frontier is informative because its approximation object is held distinct from its upper relaxation. Policy refinement improves the lower bound, but the independently fitted affine upper bound is nearly unchanged. A fixed specialized relaxation can therefore limit the attainable regret even when quadrature error is very small. This calculation is not evidence of convergence to arbitrary tolerance, and it does not establish convergence of the neural optimizer.

The classical comparator is a freshly executed direct time-control optimization. It is a serious feasible-policy and upper-bound calculation on the same original economy, but it is not two distinct state-space HJB methods and it does not furnish a matched-accuracy advantage for the neural method. A comparative claim requiring monotone PDE, semi-Lagrangian, or controlled Markov-chain baselines at the same certified accuracy remains outside the completed experiment. Likewise, the preserved Riccati-reference study used structured gain outputs; its errors cannot be relabeled as errors of the present 48-by-48 tanh actor and residual-training pipeline.

16.2. Retained comparisons and high-dimensional scope

The historical SOC-MartNet holdouts, method-specific tuning studies, and clipped-feedback comparator are preserved without suppressing unfavorable panels. The clipped-feedback comparator's lower mean cost and the absence of a robust ordering in several higher-dimensional seed panels limit what can be inferred about candidate-generation quality. They do not invalidate the stopped comparison theorem, just as the theorem does not establish empirical superiority.

The 128-dimensional experiment in the retained record certifies a rank-one-coupled Hamiltonian action oracle at frozen jets. Its nonconvex coupling can be reduced to a scalar aggregate search with strongly convex conditional subproblems. This is an action-optimization result. It is not a certified solution of a 128-dimensional dynamic economic model. The actual neural state-domain calculation in this revision concerns the two-state economy in Section 2; no action-dimension result is used to inflate that state dimension.

17. RETAINED R14: WHAT THE CERTIFIED COUNTERFACTUALS IDENTIFY

For any policy, raising k subtracts $(q - p)B(\pi)$, where

$$0 \leq B(\pi) = \mathbb{E} \int_0^\tau e^{-\rho s} \theta_s^2 / 2 \, ds \leq .02 \frac{1 - e^{-\rho}}{\rho}.$$

Consequently V_k is nonincreasing and convex. Monotonicity determines a weak sign without numerical computation, but not a strictly positive or economically well-resolved welfare loss.

TABLE XVI
CERTIFIED WELFARE LOSSES WHEN THE ADJUSTMENT PRICE RISES

p	q	Lower loss	Upper loss	Strict	Relative
0.5	2	0.01266717	0.02940793	Yes	No
2	8	0.02321063	0.04284245	Yes	Yes
4	4.1	0E-8	0.00196053	No	No
4	8	0.00433075	0.02408312	Yes	No
6	8	0E-8	0.01489694	No	No

“Strict” means a strictly positive lower endpoint. “Relative” means interval width no greater than that endpoint. A “No” in the strict column does not reverse the economic monotonicity theorem; it means that strict loss is not separated from zero by these intervals.

TABLE XVII
PRICE REGIONS IN WHICH WELFARE LOSSES ARE SEPARATED

Anchor p	Strictly positive loss	Relative width at most one
0.5	(1.093292, 8.0]	(2.617755, 8.0]
2	(2.886123, 8.0]	(6.450558, 8.0]
4	(6.207877, 8.0]	—
6	—	—

Decimal endpoints are descriptive approximations of exact rational thresholds, not proof endpoints. The exact fractions and open-endpoint convention are in the price-audit file. A dash means no such region up to price eight is established by this predicate. The relative criterion here uses the untruncated difference of value intervals and is therefore conservative.

Let $[L(k), U(k)]$ be the independently certified central-state value envelope. For $p < q$, a valid loss interval is

$$[\max\{0, L(p) - U(q)\}, \min\{U(p) - L(q), .02C_\rho(1)(q - p)\}], \quad C_\rho(h) = \frac{1 - e^{-\rho h}}{\rho}. \quad (43)$$

A positive lower endpoint establishes strict loss. We report a separate relative-resolution criterion: the width of this interval is no greater than its positive lower endpoint. Neither criterion follows automatically from a .01 bound on policy regret.

The contrast between $2 \rightarrow 8$ and $4 \rightarrow 4.1$ is substantive. The first comparison is strictly positive and relatively resolved by the current intervals. The second has a small absolute upper bound but no certified positive lower bound from interval separation. A policy may therefore be sufficiently accurate for an absolute deployment tolerance while the current value bounds remain too wide for a local comparative-static statement.

The policy library also has a direct economic interpretation. Its selector chooses a stored control vector on each exact price region; within that region no new optimization is required. The supplement lists the selected regions, their widths, consumption ranges, and average preference adjustments. These are properties of feasible chosen policies. They are not claims that the optimal policy itself is piecewise constant in the price, or that policy expenditures equal derivatives of the unknown optimum.

The tolerance .01 is an absolute loss criterion in the utility units of (5). It is not a scale-free welfare measure. The relative and strict-loss diagnostics in (43), rather than the target alone, determine which of the reported economic comparisons are resolved. No empirical calibration

or external welfare normalization is imposed to make this declared numerical target appear economically universal.

17.1. A consumption-top-up interpretation

An absolute utility tolerance is not an economic unit. We therefore define a compensation experiment before reporting it. Keep the policy's preference, wealth, adjustment cost, and stopping paths fixed, and provide an externally financed proportional consumption top-up $\lambda \geq 0$. Only the consumption entering instantaneous utility changes, from c_s to $(1 + \lambda)c_s$. This is an in-kind welfare comparison; it is not a claim that the extra consumption is financeable under the unchanged budget or the original consumption cap.

PROPOSITION 9—A scale-normalized compensation bound: *For a feasible time-control policy with $c \leq 1$, $u \in [1.2, 2.8]$ until stopping, suppose $V - J^\pi \leq \epsilon$ and $\mathbb{E} \int_0^\tau e^{-.04s} ds \geq A > 0$. If $1 - 1.8\epsilon/A > 0$, a top-up no larger than*

$$\bar{\lambda} = (1 - 1.8\epsilon/A)^{-1/1.8} - 1 \quad (44)$$

is sufficient to compensate the certified policy loss in the specified welfare experiment.

For the fresh central-state price library, the rigorous original stopping calculation supplies

$$A \geq \frac{1 - e^{-.04}}{.04} \left\{ 1 - 2 \exp\left(-\frac{.58^2}{2(.05)^2}\right) \right\}.$$

The resulting uniform top-up bound is 1.033008 percent. It is not assigned to a neural policy. Under a common positive rescaling of the entire welfare criterion, both the loss and the marginal utility gain rescale together, leaving this compensation experiment unchanged. This does not supply an empirical calibration or a wealth-equivalent claim. The exact definition prevents a numerical threshold from being presented as economically meaningful merely because it is slightly below .01.

18. BUDGET-FEASIBLE WELFARE COMPENSATION IN THE ORIGINAL ECONOMY

An absolute utility tolerance is not self-interpreting. The consumption-top-up experiment in the preceding section specifies an external financing convention, but it is not a budget-feasible wealth equivalent. We now give a sufficient initial-wealth compensation with an explicit feasible policy adjustment under the original budget and stopping rules. The object being compensated is the sharp deterministic time-control library, not the original neural policies.

Let $\pi = (c, \theta, 0)$ be the selected library policy at $(u_0, x_0) = (2, 1.25)$ and an arbitrary $k \in [.5, 8]$. Add initial wealth Δx and a constant consumption increment

$$d_c = \frac{\Delta x}{C_{.02}(1)}, \quad C_{.02}(1) = \frac{1 - e^{-.02}}{.02},$$

while retaining the preference control and zero portfolio. Before preference stopping, the wealth difference solves $\dot{z} = .02z - d_c$, $z(0) = \Delta x$, and has $z(1) = 0$ and $z(s) \geq 0$. No extra payoff, transfer after time zero, or modified consumption cap is introduced.

PROPOSITION 10—Sufficient wealth compensation: *Suppose every selected original and compensated policy avoids wealth stopping before time one, satisfies $c + d_c \leq .8$, and has preference-exit probability at most p_e . Then*

$$J_k(0, 2, 1.25 + \Delta x; \pi^+) - J_k(0, 2, 1.25; \pi) \geq d_c \cdot 8^{-1.2} C_{.04}(1)(1 - p_e). \quad (45)$$

Consequently, if the right side exceeds the original library's uniform regret bound ε_{lib} , the compensated feasible policy satisfies $J_k(0, 2, 1.25 + \Delta x; \pi^+) \geq V_k(0, 2, 1.25)$ for every library-covered price.

The comparison is sufficient compensation with an explicit policy change. It is neither the exact minimum wealth equivalent of the unchanged feedback nor a calibrated empirical welfare estimate.

Directed calculations check every one of the 20 freshly generated library policies. The preference-exit bound is below 1.207×10^{-29} . The analytic sufficient wealth increment is at most .007713143; the implementation chooses the simple rational increment $39/5000 = .0078$, or .624% of initial wealth. Its consumption increment is enclosed by [.007878259998266659, .007878259998266711]. All compensated consumption paths satisfy the original .8 cap, all terminal wealth values remain strictly above .5, and the global wealth upper bound is below $1.257958 < 2$.

The resulting uniform utility gain is at least .010094048517141695, greater than the original uniform regret upper bound .009981645275150275, with a verified slack exceeding .00011240324. An additional replay through the original stopped-payoff evaluator is retained separately. Thus the normalization is tied to a feasible economic experiment rather than to the numerical size of .01 alone. Its coverage remains the original library's central initial preference and complete price interval, not the full-state neural experiment.

19. CONCLUSION

The objective of neural Bellman operators is an accurate state-contingent solution of the original economic decision problem. The method developed here makes the connection between training and that objective explicit. A complete-cover signed residual envelope guides neural proposals, independent directed arithmetic determines acceptance, and a quantitative envelope theorem states what the accepted object certifies. Stopping accessibility corrects a genuine inconsistency of all-face hard-settlement critics without changing the economy.

The revised evidence separates numerical progress from statements that the data cannot support. The complete-cover objective produces decreasing raw certificates at all prescribed budgets in ten original-economy trajectories, reversing the earlier sampled-objective anti-alignment. Matched boundary ablations, unrestricted classical candidates, method-specific witnesses, and a computable fixed-witness floor identify the remaining obstruction. The original .01 neural target is still not met, and monotone certificate descent is not presented as monotone true policy improvement.

The nonlinear multi-product application supplies a different, positive accuracy-work result: a full-state neural initialization with a provably contracting correction attains a state-uniform total-cost bound below 1.037×10^{-6} at 128 coordinates. Strong classical alternatives remain competitive under the same certificates. A separately identified original-economy primal/dual library admits explicit budget-financed wealth compensation. These results preserve the neural-solver objective while making every theorem, numerical claim, comparison, and welfare interpretation refer to the exact object for which it is established.

APPENDIX A: PROOFS OF THE VERIFICATION AND TRANSPORT RESULTS

PROOF OF THEOREM 1: For any admissible control a , stopped Itô's formula gives

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t, z} \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_{a_s} v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

One may first stop the local martingale at a bounded localization sequence. Bounded derivatives and coefficients on the stopped compact domain give integrability, and dominated convergence then yields the displayed identity. For the candidate π , the absolute value of the right side is at most $b + \int_t^\tau e^{-\rho(s-t)} e(s) ds$. For an arbitrary original control, the residual is at most $e(s) + q(s)$, because its action is covered by the continuous maximum in (8). Hence

$$V(t, z) - v(t, z) \leq b + \int_t^\tau e^{-\rho(s-t)} \{e(s) + q(s)\} ds.$$

Adding the lower-policy error proves (9). Nonnegativity follows from admissibility of π . If an implemented action belongs to an enclosed neighborhood, the same residual and action-gap checks hold for it by interval inclusion, so the proof is unchanged. No statement about the convergence of the process that generated v or π is used. *Q.E.D.*

PROOF OF PROPOSITION 1: In the perturbed model, the candidate residual is at least $-e - d$, while the residual of any original admissible action is at most $e + q + d$. Applying the stopped identity separately in that model bounds the lower-policy and upper-value errors by the corresponding integrals. Their sum is $2e + q + 2d$, not an extra improvement-gap penalty on top of those errors. Nonnegative integrands permit replacement of the perturbed discount by ρ . The stopping-trace error is at most $b + b_{\text{par}}$ if the settlement differs by at most b_{par} . This argument neither equates the two stopping times nor assumes a pathwise coupling of different diffusions. *Q.E.D.*

APPENDIX B: PROOF OF THE JOINT STATE-PRICE CONSTRUCTION

LEMMA 1—Concavity of the covered policy functional: *For each stored policy satisfying (30), there is a concave function $F_j(z; k)$ of $z \in K$, affine in k , such that $J_k(0, z; \pi_{j,x}) \geq F_j(z; k)$. Its values at the four corners have the independently computed lower endpoints used in the state-price calculation.*

PROOF: Put $r = u - 1$ and $L = -\log c$. The determinant expression in Section 8 has the sign of

$$r^2 L^2 - 2r(r+1)L + 2(r+1) = r^2 L^2 + 2(r+1)(1 - rL) > 0,$$

while $f_{cc} < 0$. Therefore $D^2 f$ is negative definite over the stated region. In the covered Gaussian integral, the preference input is affine in initial preference and consumption is affine in initial wealth. Their image remains in the region just verified. Integrating this concave composition with positive Gaussian and time weights preserves concavity.

The full-horizon preference settlement is the concave quadratic $-.02\{(u - 2 + \int \theta)^2 + .0025\}$. By the budget shift the terminal wealth, and hence its logarithmic settlement, is independent of initial wealth. The deterministic expenditure is independent of the initial state. Subtract a uniform normal-tail and stopping allowance from this covered integral plus its terminal settlement and expenditure. The supplement proves this allowance for all $z \in K$ and $k \in [.5, 8]$. The resulting

exact function is F_j , which retains concavity and affine dependence on price. Its coefficient of $-k$ is

$$B_j^{\text{full}} = \frac{1}{2} \int_0^1 e^{-\rho s} \theta_{j,s}^2 ds.$$

At each corner the interval quadrature encloses the exact covered integral and terminal term. Its lower endpoint after the uniform corrections is no larger than F_j there. The finite quadrature rule is not assumed to reproduce the exact functional, and sampled corner values are not used without their enclosure errors. *Q.E.D.*

PROOF OF PROPOSITION 6: Let the four corners of K be z_v , and write $z = \sum_v \lambda_v(z) z_v$ using the nonnegative bilinear corner weights, which sum to one. Convexity of the verified upper object and concavity of the lower functional imply

$$V_{k_i}(0, z) \leq U_i(z) \leq \sum_v \lambda_v(z) U_i(z_v), \quad J_k(0, z; \pi_{j,x}) \geq F_j(z; k) \geq \sum_v \lambda_v(z) F_j(z_v; k).$$

For k between k_i and k_{i+1} , fixed-policy affinity in price and a common admissible policy set give convexity of V_k . Applying the value chord at the two endpoints yields the chord between the two verified upper corner combinations.

Suppose a policy lower corner endpoint is L_{jv} at k_j and its expenditure belongs to $[B_j^-, B_j^+]$. Its sign-aware lower line is

$$L_{jv} - (k - k_j) B_j^+ \quad (k \geq k_j), \quad L_{jv} - (k - k_j) B_j^- \quad (k \leq k_j).$$

Each price cell between consecutive nodes has a fixed sign for every policy. The upper chord minus this lower line is therefore an affine function $g_{jv}(k)$. For every state, the regret of policy j is at most $\sum_v \lambda_v(z) g_{jv}(k) \leq \max_v g_{jv}(k)$. Selecting a policy minimizing this last bound proves the uniform state guarantee

$$R(k) = \min_j \max_v g_{jv}(k).$$

Take all pairwise intersections of the finitely many affine functions and the price-cell endpoints. Between successive such points, their ordering does not change. The maximum for each policy and then the minimum across policies select a single affine function on that interval. Its maximum occurs at an endpoint. Exact rational evaluation at this finite set thus bounds the supremum over all real prices. A corner line is discarded only when another corner line from the same policy dominates it at both price-cell endpoints, which proves domination on the entire cell. The program records those reductions and checks the remaining intersections without a sampled-grid extrapolation. *Q.E.D.*

PROOF OF PROPOSITION 7: The independent primal and dual constructions in the supplement supply, respectively, valid original-policy lower endpoints and upper functions (29). Exact action and terminal-budget checks hold for every central policy and every shifted policy at the extremes of the initial-state rectangle. Affinity of the shift extends these feasibility checks to its interior. Lemma 1 supplies the concave lower functional with a common tail and stopping allowance. All quantities in Proposition 6 are therefore present.

The execution uses the 18 prices

$$.5, .875, 1.25, 1.625, 2, 2.5, 3, 3.5, 4, 4.25, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8.$$

The stored state–price record checks 211 candidate breakpoints and returns the exact fraction displayed following (32); upward decimal conversion gives (31). The central-state calculation uses the exact stopped-policy expenditure intervals and checks 187 candidate breakpoints with a separate all-intersections implementation. Its exact worst-case fraction is

$$\frac{1600901779824582944359922627717}{160587876576305697877131733762048},$$

whose upward display is 0.009969007711. These are computer-assisted conclusions conditional on the explicitly stated arithmetic model and analytic enclosure lemmas, not consequences of agreement between two floating-point approximations. *Q.E.D.*

APPENDIX C: ECONOMIC RESOLUTION AND THE COMMON-ENDPOINT STRESS BOUND

For $p < q$, every policy loses $(q - p)B(\pi) \geq 0$, and $B(\pi) \leq .02C_\rho(1)$. Taking suprema preserves monotonicity and gives $0 \leq V_p - V_q \leq .02C_\rho(1)(q - p)$. Combining this fact with the independently verified intervals at p and q gives (43). On any segment where the upper and lower envelope pieces are affine, both strict separation and the conservative relative-width inequality reduce to affine inequalities in q . Their roots, included with the exact envelope breakpoints, produce the resolution regions without a grid test.

Finally, if every nodal upper endpoint is enlarged by δ and every nodal lower endpoint is decreased by δ , every upper chord is enlarged by δ and every lower policy line is decreased by δ . Their maximum/minimum envelope gap increases by at most 2δ . This proves the recorded endpoint stress test and also its scope: it tests sensitivity to valid endpoint inflation, not the validity of upstream node proofs.

APPENDIX D: PROOFS FOR STOPPING ACCESSIBILITY AND CERTIFIED ITERATION

PROOF OF PROPOSITION 2: Fix $x_0 \in [1.5, 2)$ and use $c = .8$, $\theta = p = 0$. Wealth decreases, stays below two, and satisfies $x_s \geq x_0 - .8s \geq .7$ for $s \leq 1$. Thus it does not hit either wealth face. Preferences equal $2 + .05W_s$. On the event $H = \{\sup_{s \leq 1} |.05W_s| \leq .2\}$ there is no preference exit, $u \in [1.8, 2.2]$, and running utility is at least -1.5 . To see the last inequality, set $r = u - 1 \in [.8, 1.2]$. The function $-(5/4)^r/r$ is increasing on this interval since $r \log(5/4) < 1$. Its value at $.8$ exceeds -1.5 , using $(5/4)^4 < (6/5)^5$. Settlement on H is at least $-.05$, because $-.02(.2)^2 + .1 \log(.7) > -.05$. Discounting negative terms cannot worsen these lower bounds. The pathwise discounted payoff on H is therefore at least -1.55 .

Until the original stopping time, $u \in [1.2, 2.8]$. The same elementary monotonicity bounds running utility by -6 from below, and $G \geq -.1$. Even on H^c , the stopped payoff is at least $-6 - 8 - .1 = -14.1$. The Brownian maximal inequality gives $\Pr(H^c) \leq 2e^{-8}$. Hence

$$J_2(0, 2, x_0; .8, 0, 0) \geq -1.55 - 12.55(2e^{-8}).$$

Subtracting $g(0, 2, 2) = .1 \log 2 - 8$ proves (12), uniformly as $x_0 \uparrow 2$. Directed MPFR evaluation encloses its right-hand side in the interval recorded in the replication; its lower endpoint exceeds 6.37.

Finally, a smooth witness equal to g on all stopping faces and with $\sup_a \mathcal{R}_a v \leq e_+(s)$ gives $V(0, z) \leq v(0, z) + \int_0^1 e^{-.04s} e_+(s) ds$ by the stopped Itô identity. Take $z = (2, x)$, let $x \uparrow 2$, and use continuity of v and the established strict gap. The integrated uniform positive residual must exceed 6.37. A sequence with vanishing such residuals is excluded, without making any assertion that the economic control problem lacks a solution. *Q.E.D.*

PROOF OF THEOREM 2: Discounted Itô's formula, localization of the stochastic integral, and the bounded stopped-domain hypotheses give, for every admissible control a ,

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t,z}^a \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_a v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

The positive residual bound and the facewise upper discrepancy bounds imply

$$V - v \leq \int_t^T e^{-\rho(s-t)} e_+(s) ds + \sum_i \beta_i^+ \omega_i^*.$$

Apply the same identity to π with the opposite sign to obtain

$$v - J^\pi \leq \int_t^T e^{-\rho(s-t)} e_-(s) ds + \sum_i \beta_i^- \omega_i^\pi.$$

Adding proves the assertion. No common stopping time across two distinct controlled paths is assumed. For the proposed smooth hitting-probability bound w_i , optional stopping yields $\mathbb{E}[e^{-\rho(\tau-t)} w_i(\tau, Z_\tau)] \leq w_i(t, z)$; domination of the indicator on the stopping boundary then gives the stated probability bound. *Q.E.D.*

PROOF OF PROPOSITION 3: Write $Y = 2 - x > 0$ and $p = Yq$, where $q = (.15 + .65 \tanh N_p)/1.5$ is bounded. Until another exit,

$$dY = (c - .04)ds + Y\{.02 - .06xq\}ds - .2xqY dZ_s^x.$$

The bounded coefficients define a strictly positive stochastic exponential Φ . Variation of constants gives

$$Y_s = \Phi_s \left(Y_t + \int_t^s \Phi_r^{-1} (c_r - .04) dr \right) > 0,$$

since $c \geq .05$. The upper wealth face is therefore inaccessible from the interior. On the remaining state faces $B = 0$; at terminal time $1 - t = 0$. On the upper wealth face $B \geq 0$ and softplus is positive, so $v - g \geq 0$. Thus every upper-comparison trace discrepancy is zero, and the only potentially positive policy discrepancy has zero hitting probability. The same calculation applies to admissible bounded clipped head perturbations before the distance multiplication. *Q.E.D.*

PROOF OF THEOREM 3: Let $d_n = \epsilon_n/(1 - q)$. The resolvent of $I - qP_{\pi_n}$ is nonnegative and has row sums at most $(1 - q)^{-1}$, so $\|v_n - V^{\pi_n}\|_\infty \leq d_n$. Put $a_n = 2qd_n + \eta_n$. The contraction inequalities and approximate greedy inequality give

$$TV^{\pi_n} - T_{\pi_{n+1}} V^{\pi_n} \leq a_n \mathbf{1}.$$

Since $TV^{\pi_n} \geq V^{\pi_n}$, the same nonnegative resolvent gives $V^{\pi_{n+1}} \geq V^{\pi_n} - a_n \mathbf{1}/(1 - q)$. Therefore

$$\begin{aligned} V^* - V^{\pi_{n+1}} &= TV^* - T_{\pi_{n+1}} V^{\pi_{n+1}} \\ &\leq qE_n \mathbf{1} + a_n \mathbf{1} + qP_{\pi_{n+1}} (V^{\pi_n} - V^{\pi_{n+1}}) \\ &\leq \{qE_n + a_n/(1 - q)\} \mathbf{1}. \end{aligned}$$

Taking the supremum norm proves (17). The rate follows by summing the resulting geometric convolution. The comparison with a continuous-time problem is the triangle inequality after separately bounding the two transfer terms; it does not follow from the finite Markov contraction alone. *Q.E.D.*

APPENDIX E: PROOF OF THE STATE-GLOBAL INVENTORY CERTIFICATE

PROOF OF THEOREM 6: The scalar backward Riccati solutions exist on the finite horizon, remain at least .5, and are bounded above. Indeed, in remaining time their drift is $2a_i p - p^2 + q_i$, positive at .5 and negative above its positive equilibrium. The neural gains also remain at least .5 by construction. Let $d_i = \hat{p}_i - p_i^*$ and let R_i be the neural Riccati residual. Then

$$d'_i = (\hat{p}_i + p_i^* - 2a_i)d_i + R_i, \quad d_i(1) = 0.$$

The coefficient is at least $\kappa_i = 1 - 2a_i$. Integrating backward therefore yields $|d_i(t)| \leq \varepsilon_i(1 - e^{-\kappa_i(1-t)})/\kappa_i \leq D$.

The quadratic optimal value has matrix $P^* = p_\perp^*(I - M) + p_m^*M$ and a scalar noise term. Completing the square in its generator and applying Itô's formula gives the exact identity

$$\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x) = \mathbb{E} \int_0^1 |\{\hat{P}(s) - P^*(s)\}X_s|^2 ds.$$

The two projectors are orthogonal, so the integrand is bounded by $D^2|X_s|^2$. Both eigenvalues of $A - \hat{P}$ are at most $-.4$. Itô's formula for $|X_s|^2$ then gives

$$\frac{d}{ds} \mathbb{E}|X_s|^2 \leq -.8\mathbb{E}|X_s|^2 + .01d.$$

Gronwall's inequality and $|x|^2/d \leq 1$ imply $d^{-1}\mathbb{E}|X_s|^2 \leq e^{-.8s} + (.01/.8)(1 - e^{-.8s})$. Integration proves (42). The bounded deterministic gains give finite second moments and justify localization and removal of the stopping sequence in both Itô identities. No bounded-state truncation or simulated tail probability is used. *Q.E.D.*

APPENDIX F: PROOF OF THE COMPENSATION BOUND

PROOF OF PROPOSITION 9: Holding the specified paths fixed, the derivative of consumption utility with respect to the top-up is

$$\frac{\partial}{\partial \lambda} \frac{\{(1 + \lambda)c\}^{1-u}}{1 - u} = c^{1-u}(1 + \lambda)^{-u} \geq (1 + \lambda)^{-2.8}.$$

The inequality uses $c \leq 1$, $u > 1$, $u \leq 2.8$, and $\lambda \geq 0$. Integrating from zero to λ , then over the discounted alive-time measure, bounds the welfare gain below by $A\{1 - (1 + \lambda)^{-1.8}\}/1.8$. Solving for a gain of ϵ proves (44). For the time-control library, wealth feasibility excludes a wealth exit, and the retained drift bounds and Brownian maximal inequality give the displayed preference-exit probability. Surviving to terminal time alone yields the stated lower bound on discounted alive time. This argument requires no extrapolation of the certified optimal value to an altered wealth state. *Q.E.D.*

APPENDIX G: PROOFS FOR THE CERTIFICATE-ALIGNED ALGORITHM

PROOF OF THEOREM 4: For a fixed slab and sign, write $A = \max\{0, \max_i a_i\}$ and $B = \max\{0, \max_i b_i\}$. The definition of d gives $b_i \leq \max\{a_i, 0\} + d \leq A + d$, hence $B \leq A + d$. For $K = M + 1$ logits including zero, factor $e^{A/\tau}$ out of their exponential sum. The remaining sum lies in $[1, K]$. It follows that

$$A \leq \tau \log(1 + \sum_i e^{a_i/\tau}) \leq A + \tau \log K.$$

Multiply by the positive discounted slab mass, sum over the two signs and all slabs, and apply Theorem 2. This proves (20); taking $a = b$ and $d = 0$ proves (21). A nonzero independently certified trace budget can be added to both policy-loss bounds without changing the argument.

For the implementation, every stored logit is interpreted as its exact dyadic value. The maximum used to stabilize the logarithm is also an exact stored value. Directed subtraction, exponentiation, pairwise summation, logarithm, multiplication, and discounted-mass evaluation enclose the smooth objective. Directed subtraction of each nonnegative proposal logit from its retained certified endpoint encloses the correction. Its construction is valid regardless of the source of the difference between the two computational graphs. An incomplete cover would invalidate the initial use of Theorem 2; the implementation rejects that case rather than increasing a numerical tolerance. *Q.E.D.*

PROOF OF PROPOSITION 4: The initial accepted certificate is valid by independent verification. At each subsequent checkpoint, rejection leaves the incumbent unchanged, and acceptance replaces it by a valid strictly smaller bound and its associated policy. Induction proves the first assertion.

For the second assertion, consider a deterministic unit-horizon problem with discount $\rho \geq 0$, zero terminal reward, and two actions with constant rewards 0 and -1 . Put $C_\rho(h) = \int_0^h e^{-\rho s} ds$ and $v_a(t) = aC_\rho(1 - t)$. Then $v'_a - \rho v_a = -a$, so the optimal residual is $-a$ and the residual of an action with reward r is $r - a$. The signed certificate is

$$C_\rho(1)\{(-a)_+ + (a - r)_+\}.$$

The optimal action $r = 0$ paired with $a = -10$ has certificate $10C_\rho(1)$ and true regret zero. The worse action $r = -1$ paired with $a = -1$ has certificate $C_\rho(1)$ and true regret $C_\rho(1) > 0$. Both witnesses match terminal reward and both certificates are valid. Strict certificate acceptance therefore does not imply increasing policy values. This is an analytical counterexample to the inference, not a simulated observation about the original economy. *Q.E.D.*

PROOF OF PROPOSITION 5: Let $A_n = \|V_n^* - v_n\|_\infty$ and $P_n = \|V_n^\pi - v_n\|_\infty$. The evaluation and improvement inequalities imply

$$\|v_n - T_n v_{n+1}\|_\infty \leq \epsilon_n + \eta_n.$$

The Lipschitz inequalities for T_n and $T_n^{\pi_n}$ give

$$A_n \leq qA_{n+1} + \epsilon_n + \eta_n, \quad P_n \leq qP_{n+1} + \epsilon_n,$$

with $A_N, P_N \leq b$. Backward iteration and the triangle inequality give (22). Finally, $C_\rho(h) \sum_{n=0}^{N-1} e^{-\rho nh} = C_\rho(Nh)$, including its continuous extension at $\rho = 0$. The bound concerns this finite-horizon Bellman problem. Any use for a different continuous-time problem still requires a transfer theorem or a direct continuous-time certificate. *Q.E.D.*

PROOF OF PROPOSITION 8: The upper half of the verification argument gives

$$V_2(0, 2, x) \leq v(0, 2, x) + \int_0^1 e^{-.04s} e_+(s) ds$$

for every interior x . The upper trace inequalities remove any adverse settlement discrepancy. Let $x \uparrow 2$, use continuity of the fixed witness and Proposition 2, and subtract $g(0, 2, 2)$. The result is $\Delta \leq h_v + \int e^{-.04s} e_+(s) ds$. Since the integral is nonnegative, (33) follows. No policy payoff enters this lower bound; it is a statement about the residual needed to support that particular witness. Q.E.D.

APPENDIX H: PROOF FOR NONLINEAR VECTOR-STATE CONTROL

PROOF OF THEOREM 5: Let D be the cyclic first-difference matrix, so $(Dz)_i = z_i - z_{i+1}$ and $\|D\|_2 \leq 2$. Differentiating (36) gives

$$\nabla^2 \Phi(z) = \text{diag}(q_i + \frac{1}{2} \text{sech}^2 z_i) + D^\top \text{diag}(\frac{3}{10} \text{sech}^2 (Dz)_i) D.$$

All terms are positive semidefinite. Since $1 \leq q_i \leq 6/5$ and $\text{sech}^2 \leq 1$, $\nabla^2 \Phi(z) \preceq (6/5 + 1/2 + (3/10)4)I = (29/10)I$ globally.

Stack the H propagated states. Their derivative with respect to U is a block lower-triangular convolution operator B with coefficients $1, 3/5, (3/5)^2, \dots$. If S is the finite shift operator, then $B = \sum_{j=0}^{H-1} (3/5)^j S^j \otimes I_d$, so $\|B\|_2 \leq \sum_j (3/5)^j \leq 5/2$. Therefore

$$\nabla_U^2 F_x(U) = 4I + B^\top \text{diag}(\nabla^2 \Phi(z_{h+1})) B,$$

and its spectrum lies in $[4, 4 + (25/4)(29/10)] = [4, 177/8]$. This also establishes coercivity, existence, and uniqueness of the minimizer for every x .

Let $m = 4$, $L = 177/8$, and $\alpha = 2/(m + L) = 16/209$. With $g_k = \nabla F_x(U_k)$, the fundamental theorem of calculus yields

$$g_{k+1} = \left(I - \alpha \int_0^1 \nabla^2 F_x(U_k - t\alpha g_k) dt \right) g_k.$$

The integral is symmetric with spectrum in $[m, L]$. Consequently $\|g_{k+1}\|_2 \leq (L - m)/(L + m) \|g_k\|_2 = (145/209) \|g_k\|_2$. Strong convexity, followed by minimization of its quadratic lower tangent, gives $F_x(U) - \inf F_x \leq \|\nabla F_x(U)\|_2^2 / (2m) = \|\nabla F_x(U)\|_2^2 / 8$.

It remains to bound the initial gradient uniformly in the state cube and in the neural weights. The output constraint gives $|u_{0,hi}| \leq 1/4$, and $|b_{hi}| \leq .03$. State propagation therefore implies

$$|z_{hi}| \leq 1 + \frac{.25 + .03}{1 - .6} = 1.7.$$

Each coordinate of $\nabla \Phi$ has absolute value at most $(6/5)(1.7) + 1/2 + 2(3/10) = 3.14$. The backward adjoint is a geometrically weighted sum of such gradients. Hence every coordinate of $\nabla_U F_x(U_0)$ has absolute value at most $4(1/4) + 3.14/(1 - .6) = 8.85$. There are Hd coordinates, so $\|g_0\|_2^2 \leq Hd(8.85)^2$. Combining the inequalities proves (39) and solving it for k gives the work allocation.

Forward propagation, cyclic neighbor differences, and backward adjoints each require $O(Hd)$ operations; no dense Hessian need be formed. Directed verification of a stored plan replaces its

unknown exact gradient norm by an upper enclosure. It certifies that stored plan at that stored initial vector even when its floating-point optimization trajectory differs from the exact-real iteration. It does not, by itself, certify the floating-point neural/correction map simultaneously at every state. *Q.E.D.*

APPENDIX I: PROOF OF BUDGET-FEASIBLE COMPENSATION

PROOF OF PROPOSITION 10: Couple the original and compensated policies with the same preference Brownian motion. Their preference processes and adjustment costs coincide. Their portfolios are both zero. The wealth difference before stopping is

$$z(s) = \Delta x e^{.02s} - \frac{d_c}{.02} (e^{.02s} - 1) = \Delta x \frac{e^{.02} - e^{.02s}}{e^{.02} - 1} \geq 0,$$

with $z(1) = 0$. The verified wealth and consumption conditions make both policies admissible and exclude wealth stopping. Thus they have the same preference or terminal stopping time. At preference stopping, the wealth-dependent settlement increment is $.1 \log((X_s + z(s))/X_s) \geq 0$; at time one it is zero.

For any preference $u \in [1.2, 2.8]$, marginal utility is c^{-u} . Along the compensated consumption segment, $c \leq .8 < 1$, so $c^{-u} \geq .8^{-1.2}$. The mean value theorem therefore bounds the instantaneous utility increment below by $d_c .8^{-1.2}$. The survival probability at every time is at least $1 - p_e$. Integrating with discount and adding the nonnegative settlement change proves (45). The library's uniform policy-loss inequality then gives the concluding comparison with the original optimal value. The proof neither redefines that value nor asserts that the sufficient compensation is the minimum possible one. *Q.E.D.*

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